Rev. 01
February 2021

Dell EMC Unisphere for PowerMax 9.2.1

REST API Change Log

Revision: 01

Revision History

Document revision	Date	Description of changes
01	February 21	Initial release

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1 Design Overview

This document describes the changes in the 9.2 and 9.2.1 release of the public Unisphere for PowerMax REST (Representational State Transfer) API. This overview gives a concise description of the changes. Details for each of the changes are provided in the relevant resource sections. For common REST API behavior, see the Appendix. In this document “symmetrix” and “Symmetrix” is a synonym for storage array.

1.1 Performance

- Add cloud provider performance endpoints.
- Add masking view performance endpoints.
- Add new metric AvgQOSDelayPerIO to storage group category.

1.2 SLO Provisioning

- Expand editVolumeActionParam to allow enabling/disabling mobility ID on individual volumes.
- Add UUID field to storage group GET.
- Add mobility id enabled field to volume GET.
- Add snapshot policies to storage group POST.
- Add Pause/Resume/Add/Remove snapshot policy.

1.3 Replication

- Add cloud snapshot management to a storage group.
- Add MetroDR.
- Add snapshot policies.
- Add SRDFA attributes to rdf group GET.
- Add RDFG set attributes & enable DSE.
- Items that are looked up and are not found will return a 404 (Not found) response code. Inputs which are invalid will return a 400 (Bad Request) response code.

1.4 Migration

- Items that are looked up and are not found will return a 404 (Not found) response code. Inputs which are invalid will return a 400 (Bad Request) response code.

1.5 vVols

- Add VASA replication groups.

1.6 System

- Add cloud system, cloud storage groups, cloud providers, cloud jobs and cloud networking endpoints.
- Add audit log.
- Add set service level RT multiplier.
- Remove array-level job endpoints.

1.7 Updates in 9.2.1

1.7.1 Common

- Starting with the 9.2.1 release of the public Unisphere for PowerMax REST API a 503 Error Respose Code will be returned in instances where the REST API Server is unable to fulfill the user request due to a temporary server overload, resource issue, or other related issue.

1.7.2 SLO Provisioning

- Add Unreducible Data attributes to the following objects; Volume, Storage Group, SRP, Storage Group Demand Report

1.7.3 System

- Add the Rx and Tx power levels of front-end ports in mW
- Add CyberSecIQ endpoints
- Add mac_address to System Port GET
- Add Update Local Users endpoint

2 Design Details

2.1 Performance

2.1.1 cloudprovider

2.1.1.1 keys

Resource	cloudprovider
Request Type:	POST
New Endpoint Description:	Lists Cloud Providers.
Modify Endpoint Description:	Added in this Release Not Applicable Modified in this Release Not Applicable Removed in this Release Not Applicable
URL:	univmax/restapi/performance/cloudprovider/keys
User Roles:	Admin/Storage Admin/Security Admin/Monitor/Auditor/Performance Monitor
Prequisites:	Not Applicable

<table>
<tr><td>Path Parameters:</td><td>Not Applicable</td></tr>
<tr><td>Query Parameters:</td><td>Not Applicable</td></tr>
<tr><td>Request Parameters:</td><td>
<table>
<tr><td>Object</td><td colspan="3">urlParam</td></tr>
<tr><td>Name</td><td>Type</td><td>Optional</td><td>Description</td></tr>
<tr><td>symmetrixId</td><td>string</td><td>Y</td><td>SymmetrixId of a local symm</td></tr>
</table>
</td></tr>
<tr><td>Response Parameters:</td><td>
<table>
<tr><td>Object</td><td colspan="3">CloudProviderKeyResult</td></tr>
<tr><td>Name</td><td>Type</td><td>Optional</td><td>Description</td></tr>
<tr><td>cloudProviderInfos</td><td>List<CloudProviderinfo></td><td>N</td><td>Gets the value of the cloudProviderInfos property.</td></tr>
</table>
</td></tr>
</table>

2.1.1.1.1 Sample JSON Request

```
{
  "symmetrixId": "000197900256"
}
```

2.1.1.1.2 Sample JSON Response

```
{
    "cloudProviderInfo": [
        {
            "name": "minioecs2",
            "firstAvailableDate": 1565789100000,
            "lastAvailableDate": 1565841300000
        },
        {
            "name": "ECSTest",
            "firstAvailableDate": 1565789100000,
            "lastAvailableDate": 1565841300000
        }
    ]
}
```

2.1.1.2 metrics

Resource	cloudprovider
Request Type:	POST
New Endpoint Description:	Lists Cloud Providers.
Modify Endpoint Description:	Added in this Release Not Applicable Modified in this Release Not Applicable Removed in this Release Not Applicable
URL:	univmax/restapi/performance/cloudprovider/metrics
User Roles:	Not Applicable
Perquisites:	N/A
Path Parameters:	Not Applicable
Query Parameters:	N/A

Request Parameters:

Object	urlParam		
Name	Type	Optional	Description
dataFormat	DataFormat	N	Data Format (Average or Maximum).
endDate	long	N	Ending date in millisecond.
metrics	List<CloudProviderMetric>	N	Gets the value of the metrics property.
name	String	N	Cloud Provider name.
startDate	long	N	Starting date in millisecond.
symmetrixId	String	N	The Symmetrix Array ID

Response Parameters:

Object	Iterator with resultList of CloudProviderResult		
Name	Type	Optional	Description
Iterator	Iterator with resultList of CloudProviderResult	N	Gets the value of the cloudProviderInfos property.

2.1.1.2.1 Sample JSON Request

```
{
 "startDate": 1565789100000,
 "endDate": 1565841300000,
 "symmetrixId": "000197900256",
 "name": "minioecs2",
 "dataFormat": "Average",
 "metrics": [
    "DeleteResponseTime",
    "ReadResponseTime",
    "ReadThroughput",
    "TotalThroughput",
    "TotalUsedCapacity",
    "WriteResponseTime",
    "WriteThroughput"
 ]
}
```

2.1.1.2.2 Sample JSON Response

```
{
 "resultList": {
    "result": [
       {
          "DeleteResponseTime": 0.0,
          "ReadResponseTime": 0.0,
          "ReadThroughput": 0.0,
          "TotalThroughput": 0.0,
          "TotalUsedCapacity": 0.0,
```

```
            "WriteResponseTime": 0.0,
            "WriteThroughput": 0.0,
            "timestamp": 1565841000000
        },
        {
            "DeleteResponseTime": 0.0,
            "ReadResponseTime": 0.0,
            "ReadThroughput": 0.0,
            "TotalThroughput": 0.0,
            "TotalUsedCapacity": 0.0,
            "WriteResponseTime": 0.0,
            "WriteThroughput": 0.0,
            "timestamp": 1565841300000
        }
    ],
    "from": 1,
    "to": 2
},
"id": "91f111f8-8e33-4f36-8e73-89a277ccd717_0",
"count": 2,
"expirationTime": 1565872576199,
"maxPageSize": 1000
}
```


2.1.2 MaskingView

2.1.2.1 keys

Resource	MaskingView
Request Type:	POST
New Endpoint Description:	Lists Masking Views for the specified symmetrix.
Modify Endpoint Description:	Added in this Release Not Applicable Modified in this Release Not Applicable

<table>
<tr><td></td><td>Removed in this Release
Not Applicable</td></tr>
<tr><td>Perquisites:</td><td>Not Applicable</td></tr>
<tr><td>URL:</td><td>univmax/restapi/performance/MaskingView/keys</td></tr>
<tr><td>User Roles:</td><td>Not Applicable</td></tr>
<tr><td>Path Parameters:</td><td>Not Applicable</td></tr>
<tr><td>Query Parameters:</td><td>Not Applicable</td></tr>
<tr><td>Request Parameters:</td><td>
<table>
<tr><td>Object</td><td colspan="3">urlParam</td></tr>
<tr><td>Name</td><td>Type</td><td>Optional</td><td>Description</td></tr>
<tr><td>symmetrixId</td><td>string</td><td>Y</td><td>SymmetrixId of a local symm</td></tr>
</table>
</td></tr>
<tr><td>Response Parameters:</td><td>
<table>
<tr><td>Object</td><td colspan="3">CloudProviderKeyResult</td></tr>
<tr><td>Name</td><td>Type</td><td>Optional</td><td>Description</td></tr>
<tr><td>maskingViewInfos</td><td>List<MaskingViewInfo></td><td>N</td><td>Gets the value of the maskingViewInfos property.</td></tr>
</table>
</td></tr>
</table>

2.1.2.1.1 Sample JSON Request

```
{
  "symmetrixId": "000297900330"
 }
```

2.1.2.1.2 Sample JSON Response

```
{
    "maskingViewInfo": [
        {
            "maskingViewId": "Aaron",
            "firstAvailableDate": 1568644500000,
            "lastAvailableDate": 1568726400000
        },
        {
```

```
        "maskingViewId": "Aidan",
        "firstAvailableDate": 1568644500000,
        "lastAvailableDate": 1568726400000
      }
   ]
}
```


2.1.2.2 metrics

Resource	MaskingView
Request Type:	POST
New Endpoint Description:	Lists data for giving Masking View and metrics for specified time range.
Modify Endpoint Description:	Added in this Release Not Applicable Modified in this Release Not Applicable Removed in this Release Not Applicable
URL:	univmax/restapi/performance/MaskingView/metrics
User Roles:	Admin/Storage Admin/Security Admin/Monitor/Auditor/Performance Monitor
Perquisites:	Not Applicable
Path Parameters:	Not Applicable
Query Parameters:	Not Applicable

Request Parameters:

Object	urlParam		
Name	**Type**	**Optional**	**Description**
dataFormat	DataFormat	N	Data Format (Average or Maximum).
endDate	long	N	Ending date in millisecond.
metrics	List< MaskingViewMetric>	N	Gets the value of the metrics property.

	maskingViewId	String	N	Masking View ID.
	startDate	long	N	Starting date in millisecond.
	symmetrixId	String	N	The Symmetrix Array ID

Response Parameters:

Object	Iterator with resultList of MaskingViewResult		
Name	Type	Optional	Description
Iterator	Iterator with resultList of MaskingViewResult	N	Gets the value of the resultList property.

2.1.2.2.1 Sample JSON Request

```
{
  "symmetrixId": "000297900330",
  "endDate": 1568726400000,
  "dataFormat": "Average",
  "maskingViewId": "EMBEDDED_NAS_DM_MV",
  "metrics": [
        "HostIOs",
        "HostMBs",
        "ResponseTime",
        "ReadResponseTime",
        "WriteResponseTime",
        "Capacity"
        ],
  "startDate": 1568726100000
 }
```

2.1.2.2.2 Sample JSON Response

```
{
    "resultList": {
        "result": [
            {
                "HostIOs": 7.126667,
```

```
            "HostMBs": 0.11244792,
            "ResponseTime": 0.7891151,
            "ReadResponseTime": 0.0,
            "WriteResponseTime": 0.7891151,
            "Capacity": 105.82283,
            "timestamp": 1568726100000
        },
        {
            "HostIOs": 8.446667,
            "HostMBs": 0.15472656,
            "ResponseTime": 0.7947041,
            "ReadResponseTime": 0.0,
            "WriteResponseTime": 0.7947041,
            "Capacity": 105.82283,
            "timestamp": 1568726400000
        }
    ],
    "from": 1,
    "to": 2
  },
  "id": "ea92783a-7f2e-4fe2-8920-18265c746f89_0",
  "count": 2,
  "expirationTime": 1568728713840,
  "maxPageSize": 1000
}
```


2.1.3 StorageGroup

Resource	StorageGroup
Request Type:	POST
New Endpoint Description:	Lists data for specified storage group and metrics for given time range.
Modify Endpoint Description:	Added in this Release Not Applicable Modified in this Release

	Not Applicable
	Removed in this Release
	Not Applicable
URL:	univmax/restapi/performance/StorageGroup/metrics
User Roles:	Admin/Storage Admin/Security Admin/Monitor/Auditor/Performance Monitor
Perquisites:	Not Applicable
Path Parameters:	Not Applicable
Query Parameters:	Not Applicable

Request Parameters:

Object	urlParam		
Name	Type	Optional	Description
dataFormat	DataFormat	N	Data Format (Average or Maximum).
endDate	long	N	Ending date in millisecond.
metrics	List< StorageGroupMetric >	N	Gets the value of the metrics property.
storageGroupId	String	N	Gets the value of the storageGroupId property.
startDate	long	N	Starting date in millisecond.
symmetrixId	String	N	The Symmetrix Array ID

Response Parameters:

Object	Iterator with resultList of StorageGroupResult		
Name	Type	Optional	Description
Iterator	Iterator with resultList of StorageGroupResult	N	Gets the value of the resultList property.

2.1.3.1 Sample JSON Request

```
{
  "symmetrixId": "000196700153",
  "endDate": 1567161900000,
  "dataFormat": "Average",
```

```
  "storageGroupId": "irco1sd10",
  "metrics": [
        "AvgQOSDelayPerIO"
  ],
  "startDate": 1567161600000
}
```

2.1.3.2 Sample JSON Response

```
{
    "resultList": {
        "result": [
            {
                "AvgQOSDelayPerIO": 0.0,
            },
            {
                " AvgQOSDelayPerIO": 0.0
            }
        ],
        "from": 1,
        "to": 2
    },
    "id": "ea92783a-7f2e-4fe2-8920-18265c746f89_0",
    "count": 2,
    "expirationTime": 1568728713840,
    "maxPageSize": 1000
}
```

2.1.1 SRP

Resource	SRP
Request Type:	POST
New Endpoint Description:	Lists data for specified SRP and metrics for given time range.

Modify Endpoint Description:

Added in this Release
Not Applicable
Modified in this Release
Not Applicable
Removed in this Release
Not Applicable

URL: univmax/restapi/performance/SRP/metrics

User Roles: Admin/Storage Admin/Security Admin/Monitor/Auditor/Performance Monitor

Perquisites: Not Applicable

Path Parameters: Not Applicable

Query Parameters: Not Applicable

Request Parameters:

Object	urlParam		
Name	Type	Optional	Description
dataFormat	DataFormat	N	Data Format (Average or Maximum).
endDate	long	N	Ending date in millisecond.
metrics	List< SRPMetric >	N	Gets the value of the metrics property.
srpId	String	N	Gets the value of the srpId property.
startDate	long	N	Starting date in millisecond.
symmetrixId	String	N	The Symmetrix Array ID

Response Parameters:

Object	Iterator with resultList of SRPResult		
Name	Type	Optional	Description
Iterator	Iterator with resultList of SRPResult	N	Gets the value of the resultList property.

2.1.1.1 Sample JSON Request


```
{
  "symmetrixId": "000196700153",
  "endDate": 1571322300000,
  "dataFormat": "Average",
  "srpId": "SRP_1",
  "metrics": [
        "TotalSnapshotAllocatedCapacity"
  ],
  "startDate": 1571321400000
}
```


2.1.1.2 Sample JSON Response


```
{
    "resultList": {
        "result": [
            {
                "TotalSnapshotAllocatedCapacity”:0.0,
               "timestamp": 1571321400000
            },
            {
                "TotalSnapshotAllocatedCapacity": 0.0,
                 "timestamp": 1571321700000
            }
        ],
        "from": 1,
        "to": 2
    },
    "id": "ea92783a-7f2e-4fe2-8920-18265c746f89_0",
    "count": 2,
    "expirationTime": 1571653815290,
    "maxPageSize": 1000
}
```

2.2 SLO Provisioning

2.2.1 volume

2.2.1.1 PUT

<table>
<tr><td>Resource</td><td>volume</td></tr>
<tr><td>Request Type:</td><td>PUT</td></tr>
<tr><td>New Endpoint Description:</td><td>Not Applicable</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><td>Added in this Release</td></tr>
<tr><td>Enable/Disable Mobility ID on individual volumes</td></tr>
<tr><td>Modified in this Release</td></tr>
<tr><td>Not Applicable</td></tr>
<tr><td>Removed in this Release</td></tr>
<tr><td>Not Applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>… /92/sloprovisioning/symmetrix/{symmetrixId}/volume/{volumeId}</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>Not Applicable</td></tr>
<tr><td>Query Parameters:</td><td>Not applicable</td></tr>
<tr><td>Request Parameters:</td><td>SYNCHRONOUS:
<table>
<tr><td>Object</td><td colspan="3">editVolumeActionParam</td></tr>
<tr><td>Attribute Name</td><td>Type</td><td>Optional</td><td>Description</td></tr>
<tr><td>enable_mobility_id_param</td><td>EnableMobilityIdParamType</td><td>Yes</td><td>Enable mobility Id on volume</td></tr>
<tr><td>Object</td><td colspan="3">enable_mobility_id_param</td></tr>
</table>
</td></tr>
</table>

	Attribute Name	Type	Optional	Description
	enable_mobility_id	boolean	No	Enable/Disable mobility Id on volume
Response Parameters:				

2.2.1.1.1 Sample JSON request

```
{
        "editVolumeActionParam": {
                "enable_mobility_id_param": {
                        "enable_mobility_id": false
                }
        }
}
```

2.2.1.2 GET (List)

Resource	volume
Request Type:	GET
New Endpoint Description:	Not Applicable
Modify Endpoint Description:	Added in this Release Query parameters added: has_effective_wwn effective_wwn Modified in this Release Not Applicable Removed in this Release Not Applicable
URL:	… /92/sloprovisioning/symmetrix/{symmetrixId}/volume?{attribute=attributeValue}
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable

<table>
<tr><td>Path Parameters:</td><td><table>
<tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>Unique identifier of the symmetrix</td></tr>
</table></td></tr>
<tr><td>Query Parameters:</td><td><table>
<tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr>
<tr><td>has_effective_wwn</td><td>String</td><td>An optional value that filters the returned list so it displays volume ids that have effective wwns (true/false)
"has_effective_wwn=true"</td></tr>
<tr><td>effective_wwn</td><td>String</td><td>An optional value that filters the returned list so it displays volume ids that contain the specified effective_wwn
example: “effective_wwn =1234567890123456789012345678901234 5”</td></tr>
</table></td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>Not Applicable</td></tr>
</table>

2.2.1.2.1 Sample JSON request


```
{
   "resultList": {
      "result": [
         {
            "volumeId": "00307"
         },
         {
            "volumeId": "00308"
         },
         {
            "volumeId": "00309"
         },
         {
            "volumeId": "0030A"
         },
         {
            "volumeId": "0030B"
         },
         {
            "volumeId": "0030C"
         }
      ],
      "from": 1,
      "to": 6
   },
   "id": "48803eaf-73e1-42b6-92e3-bb766768966d_0",
   "count": 6,
   "expirationTime": 1575029788873,
   "maxPageSize": 1000
}
```

2.2.1.1 GET (Object)

<table>
<tr><td>Resource</td><td>volume</td></tr>
<tr><td>Request Type:</td><td>GET</td></tr>
<tr><td>New Endpoint Description:</td><td>Not Applicable</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><td>Added in this Release</td></tr>
<tr><td>mobility_id_enabled</td></tr>
<tr><td>Modified in this Release</td></tr>
<tr><td>Not Applicable</td></tr>
<tr><td>Removed in this Release</td></tr>
<tr><td>Not Applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>… /92/sloprovisioning/symmetrix/{symmetrixId}/volume/{volumeId}</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>Not Applicable</td></tr>
<tr><td>Query Parameters:</td><td>Not Applicable</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>
<table>
<tr><td>Object</td><td colspan="3">Volume</td></tr>
<tr><td>Attribute Name</td><td>Type</td><td>Optional</td><td>Description</td></tr>
<tr><td>mobility_id_enabled</td><td>boolean</td><td>No</td><td>Mobility/Compatability ID</td></tr>
</table>
</td></tr>
</table>

2.2.1.1.1 Sample JSON response

```
{
   “mobility_id_enabled”: true
}
```

2.2.2 srp

2.2.2.1 GET (Object)

Resource	srp
Request Type:	GET

<table>
<tr><td>New Endpoint Description:</td><td>Not Applicable</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Added attribute
service_levels</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not Applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not Applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>… /92/sloprovisioning/symmetrix/{symmetrixId}/srp/{srpId}</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>Unique identifier of the symmetrix</td></tr>
<tr><td>srpId</td><td>String</td><td>Unique identifier of the SRP</td></tr>
</table>
</td></tr>
<tr><td>Query Parameters:</td><td>Not Applicable</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>
<table>
<tr><th>Object</th><th colspan="3">SrpType</th></tr>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>service_levels</td><td>Array[String]</td><td>Yes</td><td>A List of Servce levels for the specified SRP</td></tr>
</table>
</td></tr>
</table>

2.2.2.1.1 Sample JSON Response

```
{
  "service_levels": [
    "Bronze",
    "Diamond",
    "Gold",
    "None",
    "Optimized",
    "Platinum",
    "Silver"
  ]
```

```
}
```

2.2.3 host

2.2.3.1 GET (Object)

<table>
<tr><td>Resource</td><td>host</td></tr>
<tr><td>Request Type:</td><td>GET</td></tr>
<tr><td>New Endpoint Description:</td><td>Not Applicable</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Added attribute
bw_limit</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not Applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not Applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>… /92/sloprovisioning/symmetrix/{symmetrixId}/host/{hostId}</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>Unique identifier of the symmetrix</td></tr>
<tr><td>hostId</td><td>String</td><td>Unique identifier of the Host</td></tr>
</table>
</td></tr>
<tr><td>Query Parameters:</td><td>Not Applicable</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>
<table>
<tr><th>Object</th><th colspan="3">HostType</th></tr>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>bw_limit</td><td>long</td><td>Yes</td><td>The Bandwidth limit of the Host</td></tr>
</table>
</td></tr>
</table>

2.2.3.1.1 Sample JSON Response


```
{
  "bw_limit":100
}
```


2.2.4 initiator

2.2.4.1 GET (Object)

Resource	initiator
Request Type:	GET
New Endpoint Description:	Not Applicable
Modify Endpoint Description:	Added in this Release Added attribute • fcid Modified in this Release Not Applicable Removed in this Release Not Applicable
URL:	… /92/sloprovisioning/symmetrix/{symmetrixId}/initiator/{initiatorId}
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable
Path Parameters:	Attribute Name / Type / Description symmetrixId / String / Unique identifier of the symmetrix initiatorId / String / Unique identifier of the initiator
Query Parameters:	Not Applicable
Request Parameters:	Not applicable

Response Parameters:	Object	Details of the specified initiator		
	Attribute Name	Type	Optional	Description
	fcid	String	No	FCID value

Sample JSON Response


```
{
   "initiatorId": "50000973b0040045",
   "symmetrixPortKey": [
      {
         "directorId": "FA-1D",
         "portId": "5"
      }
   ],
   "type": "Fibre",
   "fcid": "22a00",
   "logged_in": true,
   "on_fabric": true,
   "fabric_name": "1152921676278634649",
   "flags_in_effect": "Common_Serial_Number(C), SCSI_3(SC3), SPC2_Protocol_Version(SPC2), SCSI_Support1(OS2007)",
   "num_of_vols": 0,
   "num_of_host_groups": 0,
   "num_of_masking_views": 0,
   "num_of_powerpath_hosts": 0
}
```


2.2.4.2 PUT

Resource	initiator
Request Type:	PUT
New Endpoint Description:	Not Applicable
Modify Endpoint Description:	Added in this Release • removeMaskingEntry now returns a 204 if successful instead of a 500 and error, unless it's the last initiator path(director:port) for that HBA, in which case it will still return a 500 and an error due to a bug that will be addressed in the next release. Modified in this Release Not Applicable

<table>
<tr><td></td><td><table><tr><th>Removed in this Release</th></tr><tr><td>Not Applicable</td></tr></table></td></tr>
<tr><td>URL:</td><td>… /92/sloprovisioning/symmetrix/{symmetrixId}/initiator/{initiatorId}</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td><table><tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr><tr><td>symmetrixId</td><td>String</td><td>Unique identifier of the symmetrix</td></tr><tr><td>initiatorId</td><td>String</td><td>The Initiator id</td></tr></table></td></tr>
<tr><td>Query Parameters:</td><td>Not applicable</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>Not Applicable</td></tr>
</table>

2.2.5 storagegroup

2.2.5.1 POST

<table>
<tr><td>Resource</td><td>storagegroup</td></tr>
<tr><td>Request Type:</td><td>POST</td></tr>
<tr><td>New Endpoint Description:</td><td>Not Applicable</td></tr>
<tr><td>Modify Endpoint Description:</td><td><table><tr><th>Added in this Release</th></tr><tr><td>Associate snapshot policies to storage group on create</td></tr><tr><th>Modified in this Release</th></tr><tr><td>Not Applicable</td></tr><tr><th>Removed in this Release</th></tr><tr><td>Not Applicable</td></tr></table></td></tr>
<tr><td>URL:</td><td>… /92/sloprovisioning/symmetrix/{symmetrixId}/storagegroup</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator</td></tr>
</table>

Prerequisite: Not applicable

Path Parameters:

Attribute Name	Type	Description
symmetrixId	String	Unique identifier of the symmetrix

Query Parameters: Not applicable

Request Parameters:

Object	createStorageGroupParam		
Attribute Name	Type	Optional	Description
snapshot_policies	SnapshotProtectionPolicyIDType	Yes	Array of snapshot policy ids to associate to a parent or standalone storage group

Object	SloBasedStorageGroupParamType		
Attribute Name	Type	Optional	Description
snapshot_policies	SnapshotProtectionPolicyIDType	Yes	Array of snapshot policy ids to associate to a child storage group

Response Parameters: Not Applicable

2.2.5.1.1.1 Sample JSON response

```
{
  "storageGroupId": "sg_id",
  "snapshot_policy": [
    "policy_id"
  ]

}
```

2.2.5.2 PUT

<table>
<tr><td>Resource</td><td>storagegroup</td></tr>
<tr><td>Request Type:</td><td>PUT</td></tr>
<tr><td>New Endpoint Description:</td><td>Not Applicable</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Associate snapshot policy
Disassociate snapshot policy
Resume snapshot policy
Pause snapshot policy</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>… /92/sloprovisioning/symmetrix/{symmetrixId}/storagegroup/{storageGroupID}</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>Unique identifier of the symmetrix</td></tr>
<tr><td>storageGroupId</td><td>String</td><td>The Storage Group Name</td></tr>
</table>
</td></tr>
<tr><td>Query Parameters:</td><td>Not applicable</td></tr>
<tr><td>Request Parameters:</td><td>
<table>
<tr><th>Object</th><th colspan="3">EditStorageGroupParamType</th></tr>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
</table>
</td></tr>
</table>

edit_snapshot_policies_param	EditSnapshotProtectionPoliciesParamType	Yes	Allows editing of the snapshot protection policies on a storage group

Object	EditSnapshotProtectionPoliciesParamType		
Attribute Name	Type	Optional	Description
resume_snapshot_policy_param	ResumeSnapshotProtectionPolicyParamType	Yes	Resumes the specified snapsho policy on the storage group
suspend_snapshot_policy_param	SuspendSnapshotProtectionPolicyParamType	Yes	Suspends the specified snapsho policy on the storage group
disassociate_snapshot_policy_param	DisassociateSnapshotProtectionPolicyParamType	Yes	Disassociates the specified snapsho policy on the storage group
associate_snapshot_policy_param	AssociateSnapshotProtectionPolicyParamType	Yes	Associates the specified snapsho policy(s) to the storage group

Object	ResumeSnapshotProtectionPolicyParamType		
Attribute Name	Type	Optional	Description
snapshot_policies	SnapshotProtectionPolicyIDType	No	Array of snapsho policy ids

Object	SuspendSnapshotProtectionPolicyParamType		
Attribute Name	Type	Optional	Description

	snapshot_policies	SnapshotProtectionPolicyIDType	No	Array of snapshot policy ids

Object	DisassociateSnapshotProtectionPolicyParamType		
Attribute Name	Type	Optional	Description
snapshot_policies	SnapshotProtectionPolicyIDType	No	Array of snapshot policy ids

Object	AssociateSnapshotProtectionPolicyParamType		
Attribute Name	Type	Optional	Description
snapshot_policies	SnapshotProtectionPolicyIDType	No	Array of snapshot policy ids

Response Parameters:

SYNCHRONOUS:

Object	storageGroup		
Attribute Name	Type	Optional	Description
"snapshot_policies": ["WeeklyDefault"],			

2.2.5.2.1 Sample JSON Response

```
{
   "editStorageGroupActionParam": {
      "edit_snapshot_policies_param": {
         "pause_snapshot_policy_param": {
            "snapshot_policies": [
               "policy_id"
            ]
         }
      }
   }
}

{
   "editStorageGroupActionParam": {
      "edit_snapshot_policies_param": {
         "resume_snapshot_policy_param": {
            "snapshot_policies": [
               "policy_id"
            ]
         }
      }
```

```
   }
}

{
   "editStorageGroupActionParam": {
      "edit_snapshot_policies_param": {
         "disassociate_snapshot_policy_param": {
            "snapshot_policies": [
               "policy_id"
            ]
         }
      }
   }
}

{
   "editStorageGroupActionParam": {
      "edit_snapshot_policies_param": {
         "associate_snapshot_policy_param": {
            "snapshot_policies": [
               "policy_id"
            ]
         }
      }
   }
}
```

2.2.5.3 GET (Object)

Resource	storagegroup
Request Type:	GET
New Endpoint Description:	Not Applicable
Modify Endpoint Description:	Added in this Release Added UUID Field Modified in this Release Not Applicable Removed in this Release Not Applicable
URL:	… /92/sloprovisioning/symmetrix/{symmetrixId}/storagegroup/{storagegroupId}
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor

<table>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Query Parameters:</td><td>Not applicable</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>
<table>
<tr><td>Object</td><td colspan="3">storageGroup</td></tr>
<tr><td>Attribute Name</td><td>Type</td><td>Optional</td><td>Description</td></tr>
<tr><td>uuid</td><td>String</td><td>yes</td><td>Unique identifier of the storage group instance</td></tr>
</table>
</td></tr>
</table>

2.2.5.3.1 Sample JSON Request

```
{
   "storageGroupId": "0ShaneLocalR2",
   "slo": "Diamond",
   "service_level": "Diamond",
   "base_slo_name": "Diamond",
   "srp": "SRP_1",
   "slo_compliance": "NONE",
   "num_of_vols": 4,
   "num_of_child_sgs": 0,
   "num_of_parent_sgs": 0,
   "num_of_masking_views": 0,
   "num_of_snapshots": 0,
   "cap_gb": 8.0,
   "device_emulation": "FBA",
   "type": "Standalone",
   "unprotected": true,
   "compression": true,
   "vp_saved_percent": 100.0
   "uuid": 00017
}
```

2.3 Replication

- Items that are looked up and are not found will return a 404 response code. Inputs which are invalid will return a 400 response code.

2.3.1 cloudsnapshot

2.3.1.1 GET (List)

<table>
<tr><td>Resource</td><td>cloudsnapshot</td></tr>
<tr><td>Request Type:</td><td>GET</td></tr>
<tr><td>New Endpoint Description:</td><td>Gets a list of Cloud snapshots for a specified Local Storage Group</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Not Applicable</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not Applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not Applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>…/92/replication/symmetrix/{symmetrixId}/storagegroup/{storageGroupId}/cloudsnapshot/</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>No</td><td>Unique identifier of the symmetrix</td></tr>
<tr><td>storageGroupId</td><td>String</td><td>No</td><td>The Storage Group name</td></tr>
</table>
</td></tr>
<tr><td>Query Parameters:</td><td>See GET (List)</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>See GET (List)</td></tr>
</table>

2.3.1.1.1 Sample JSON response


```
{
  "cloud_snapshot_infos": [
   {
     "cloud_snapshot_id": "124587888",
     "snapshot_name": "Weekly_123",
     "creation_date": "Wed, 05 Dec 2018 10:14:46",
     "creation_date_timestamp": 1544004886
```

```
  },
  {
     "cloud_snapshot_id": "124587889",
     "snapshot_name": "Weekly_123",
     "creation_date": "Wed, 12 Dec 2018 10:14:46",
     "creation_date_timestamp": 1544609686
  }
}
```

2.3.1.2 GET (Object)

Resource	cloudsnapshot
Request Type:	GET
New Endpoint Description:	Gets a Cloud Snapshot details
Modify Endpoint Description:	Added in this Release Not Applicable Modified in this Release Not Applicable Removed in this Release Not Applicable
URL:	…/92/replication/symmetrix/{symmetrixId}/storagegroup/{storageGroupId}/cloudsnapshot/{cloud_snapshot_id}
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable
Path Parameters:	Attribute Name / Type / Optional / Description symmetrixId / String / No / Unique identifier of the symmetrix storageGroupId / String / No / The Storage Group name cloud_snapshot_id / String / No / The Cloud Snapshot Id (Not name)
Query Parameters:	Not applicable
Request Parameters:	Not applicable

Response Parameters:	See GET (Object)

2.3.1.2.1 Sample JSON response

See GET (Object)

2.3.1.3 POST

Resource	cloudsnapshot
Request Type:	POST
New Endpoint Description:	Create a Cloud Snapshot for a local Storage Group
Modify Endpoint Description:	Added in this Release Not Applicable Modified in this Release Not Applicable Removed in this Release Not Applicable
URL:	…/92/replication/symmetrix/{symmetrixId}/storagegroup/{storageGroupId}/cloudsnapshot
User Roles:	Administrator/Storage Administrator
Prerequisite:	Not applicable
Path Parameters:	Attribute Name / Type / Optional / Description symmetrixId / String / No / Unique identifier of the symmetrix storageGroupId / String / No / The Storage Group name

Query Parameters: Not applicable

Request Parameters:

Object	CreateCloudSnapshotParamType		
Attribute Name	Type	Optional	Description
cloud_provider_id	String	No	The name of the cloud provider where the snapshot will be archived to
snapshot_name	String	No	The Snapshot name that is used to group cloud snapshots
expiry_time_days	int	Yes	The time that the snapshot generation is to live for in Days. A minimum of 7 days must be applied in either days, hours or both

Response Parameters:

SYNCHRONOUS:

Object	cloudSnapshot		
Attribute Name	Type	Optional	Description
<Cloud Snapshot> is detailed in GET (Object)			

ASYNCHRONOUS:

Object	job

	Attribute Name	Type	Optional	Description
	<Job> is detailed in the Appendix.			

2.3.1.3.1 Sample JSON request

```
{
  "cloud_provider_id": "Amazon",
  "snapshot_name": "Weekly_123",
  "expiry_time_days": 100
}
```

2.3.1.4 PUT

Resource	cloudsnapshot
Request Type:	PUT
New Endpoint Description	Modify a Cloud Snapshot
Modify Endpoint Description:	Added in this Release Not Applicable Modified in this Release Not Applicable Removed in this Release Not Applicable
URL:	…/92/replication/symmetrix/{symmetrixId}/storagegroup/{storageGroupId}/cloudsnapshot/{cloud_snapshot_id}
User Roles:	Administrator/Storage Administrator
Prerequisite:	Not applicable

Path Parameters:

Attribute Name	Type	Optional	Description
symmetrixId	String	No	Unique identifier of the symmetrix
storageGroupId	String	No	The Storage Group name
cloud_snapshot_id	String	No	The Cloud Snapshot Id (Not name)

Query Parameters:

Not applicable

Request Parameters:

See PUT

Response Parameters:

SYNCHRONOUS:

Object	cloudSnapshot		
Attribute Name	Type	Optional	Description
<Cloud Snapshot> is detailed in GET (Object)			

ASYNCHRONOUS:

Object	job		
Attribute Name	Type	Optional	Description
<Insert the noun or noun phrase that describes what is detailed> is detailed in the Common Objects section.			

2.3.1.4.1 Sample JSON request

See PUT

2.3.1.5 DELETE

Resource	cloudsnapshot
Request Type:	DELETE
New Endpoint Description:	Not Applicable
Modify Endpoint	

<table>
<tr><td>Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Deletes a Cloud Snapshot when in the following states only:
Created
Queued
Archived</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not Applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not Applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>…/92/replication/symmetrix/{symmetrixId}/storagegroup/{storageGroupId}/cloudsnapshot/{cloud_snapshot_id}</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>No</td><td>Unique identifier of the symmetrix</td></tr>
<tr><td>storageGroupId</td><td>String</td><td>No</td><td>The Storage Group name</td></tr>
<tr><td>cloud_snapshot_id</td><td>String</td><td>No</td><td>The Cloud Snapshot Id (Not name)</td></tr>
</table>
</td></tr>
<tr><td>Query Parameters:</td><td>Not applicable</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>No content</td></tr>
</table>

2.3.2 metrodr

2.3.2.1 GET (List)

<table>
<tr><td>Resource</td><td>metrodr</td></tr>
<tr><td>Request Type:</td><td>GET</td></tr>
<tr><td>New Endpoint Description:</td><td>Ability to view list of MetroDR Environment names</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>…/92/replication/symmetrix/{symmetrixId}/metrodr</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>No</td><td>Unique identifier of the symmetrix</td></tr>
</table>
</td></tr>
<tr><td>Query Parameters:</td><td>Not applicable</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>
<table>
<tr><th>Object</th><th colspan="3">MetroDREnvironmentList</th></tr>
<tr><th>Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>names</td><td>List</td><td>No</td><td>List of Metro DR Environment Names</td></tr>
</table>
</td></tr>
</table>

2.3.2.1.1 Sample JSON response

```
{
   "names": [ “Name1”, “Name2”]
}
```

2.3.2.2 GET (Object)

<table>
<tr><td>Resource</td><td>metrodr</td></tr>
<tr><td>Request Type:</td><td>GET</td></tr>
<tr><td>New Endpoint Description:</td><td>Ability to view MetroDR Environment details</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>…/92/replication/symmetrix/{symmetrixId}/metrodr/{environmentName}</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>No</td><td>Unique identifier of the symmetrix</td></tr>
<tr><td>environmentName</td><td>String</td><td>No</td><td>The Name of the Environment</td></tr>
</table>
</td></tr>
<tr><td>Query Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr>
<tr><td>config</td><td>boolean</td><td>An optional value that dispalys additional SRDF Group and pair configuration information for the environment.</td></tr>
</table>
</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>
<table>
<tr><th>Object</th><th colspan="3">MetroDrEnvironment</th></tr>
<tr><th>Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>name</td><td>String</td><td>No</td><td>The name of the MetroDR Environment</td></tr>
</table>
</td></tr>
</table>

environment_state	String	Yes	The state of the environment
valid	Boolean	No	Indicates that the MetroDR Environment is Valid.
metro_state	String	No	The State of the Metro Session in the Environment.
metro_link_state	String	No	The state of the metro link in the environment
metro_exempt	Boolean	No	Some volumes may be exempt in the Metro session
metro_service_state	String	No	The Service State of the Metro session.
metro_witness_state	String	No	The state of the Metro witness
metro_percent_complete	Integer	No	The percent copied value
metro_remain_capacity_to_copy_mb	Double	No	The capacity remaining to copy (in MB)
metro_r1_connectivity_health	String	No	The Connectivity Health for the Metro R1 array
metro_r2_connectivity_health	String	No	The Connectivity Health for the Metro R2 array
metro_r1_array_health	String	No	The Array Health for the Metro R1 array
metro_r2_array_health	String	No	The Array Health for the Metro R2 array

dr_state	String	No	The State of the DR Session in the Environment.
dr_link_state	String	No	The state of the DR link in the environment
capacity_gb	Double	No	The capacity, in GB, of the MetroDR Environment.
dr_exempt	Boolean	No	Some devices may be exempt in the DR session
dr_service_state	String	No	The Service State of the DR leg.
environment_exempt	Boolean	No	Indicates that there are Exempt devices in the Environment
dr_remain_capacity_to_copy_mb	Integer	No	The capacity remaining to copy (in MB)
dr_percent_complete	Double	No	The percent copied value
dr_rdf_mode	String	No	Indicates the mode for the DR session
Configuration	MetroDrEnvironmentConfig	Yes	The configuration of the Metro DR environment

Object	MetroDrEnvironmentConfig		
Name	Type	Optional	Description
metro_r1_array	String	No	The serial number for the Metro R1 array
metro_r1_metro_r2_rdfg	Integer	No	The SRDF group number on the Metro R1 array to the Metro R2 array

metro_r1_dr_rdfg	Integer	No	The SRDF group number on the Metro R1 array to the DR array
metro_r2_array	String	No	The serial number for the Metro R2 array
metro_r2_metro_r1_rdfg	Integer	No	The SRDF group number on the Metro R2 array to the Metro R1 array
metro_r2_dr_rdfg	Integer	No	The SRDF group number on the Metro R2 array to the DR array
dr_array	String	No	The serial number for the DR array
dr_metro_r1_rdfg	Integer	No	The SRDF group number on the DR array to the Metro R1 array
dr_metro_r2_rdfg	Integer	No	The SRDF group number on the DR array to the Metro R2 array
metro_dr_volume_pairing	MetroDrTriangle	No	The collection of volume relationships in the environment

Object	MetroDrTriangle		
Name	Type	Optional	Description
metro_r1_volume_name	String	No	The name of the volume in the R1 role of the Metro session
metro_r2_volume_name	String	No	The name of the volume in the R2 role of the Metro session
dr_volume_name	String	No	The name of the volume in the DR role
metro_r1_r2_paired	Boolean	No	Are the R1 and R2 volumes paired in the Metro session
metro_r1_dr_paired	Boolean	No	Are the metro R1 and DR volumes paired in the environment
metro_r2_dr_paired	Boolean	No	Are the metro R2 and DR volumes paired in the environment
metro_r1_mapped	Boolean	No	Is the Metro R1 volume mapped
metro_r2_mapped	Boolean	No	Is the Metro R2 volume mapped
dr_mapped	Boolean	No	Is the DR volume mapped

2.3.2.2.1 Sample JSON response


```
{
   "name": "PF_MetroDR",
   "valid": true,
   "environment_state": "Normal",
   "capacity_gb": 3.005,
   "metro_state": "ActiveActive",
   "metro_link_state": "Online",
   "metro_service_state": "Active_HA",
   "metro_witness_state": "Available",
   "metro_percent_complete": 100,
   "metro_r1_connectivity_health": "Normal",
   "metro_r1_array_health": "Normal",
   "metro_r2_connectivity_health": "Normal",
   "metro_r2_array_health": "Normal",
   "dr_state": "Suspended",
```

```
    "dr_link_state": "Online",
    "dr_service_state": "Inactive",
    "dr_rdf_mode": "Asynchronous"
}
```


2.3.2.3 PUT

Resource	metrodr
Request Type:	PUT
New Endpoint Description:	Modifies the MetroDr Environment: • Ability to establish a session in an SRDF/MetroDR Environment • Ability to restore a session in an SRDF/MetroDR Environment • Ability to restore a session in an SRDF/MetroDR Environment • Ability to split the dr session within a MetroDR Environment • Ability to Recover the MetroDR Environment • Ability to Failover the MetroDR Environment • Ability to Failback the MetroDR Environment • Ability to Update R1 the dr session in the MetroDR Environment • Ability to Set Mode the DR session in the MetroDR Environment
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL:	…/92/replication/symmetrix/{symmetrixId}/metrodr/{environmentName}
User Roles:	Administrator/Storage Administrator
Prerequisite:	Not applicable
Path Parameters:	Attribute Name | Type | Optional | Description

	symmetrixId	String	No	Unique identifier of the symmetrix	
	environmentName	String	No	The Name of the Environment	

Query Parameters: Not applicable

Request Parameters:

Object	SrdfMetroDREnvironmentUpdate		
Name	Type	Optional	Description
action	MetroDREnvironmentAction	No	The action to be performed
establish	MetroDREnvEstablishParam	No	• metro • dr • force • symForce
suspend	MetroDREnvSuspendParam	No	• metro • dr • keepR1 • keepR2 • force • symForce
restore	MetroDREnvRestoreParam	No	• metro • dr • force • symForce
split	MetroDREnvSplitParam	No	• force • symForce
recover	MetroDREnvRecoverParam	No	• force • symForce
failover	MetroDREnvFailoverParam	No	• force • symForce
failback	MetroDREnvFailbackParam	No	• force • symForce
updateR1	MetroDREnv UpdateR1Param	No	• force • symForce
setMode	MetroDRSetModeParam	No	• Force • mode • symForce

Object	MetroDREnvironmentAction		
Attribute Name	Type	Optional	Description
Establish	String	NO	Establish a session within the environment
Suspend	String	NO	Suspend a session

			within the Environment
Restore	String	NO	Restore a session within the Environment
Split	String	NO	Split the DR Session
Recover	String	NO	Recover the MetroDR Environment
Failover	String	NO	Failover the DR session
Failback	String	NO	Failback the DR session
UpdateR1	String	NO	Update R1 the DR session
SetMode	String	NO	Set Mode on the DR session

Response Parameters:

SYNCHRONOUS:

Object	MetroDrEnvironment		
Attribute Name	Type	Optional	Description
< MetroDrEnvironment > is detailed in GET (Object)			

ASYNCHRONOUS:

Object	job		
Attribute Name	Type	Optional	Description
<Job> is detailed in the Appendix.			

2.3.2.3.1 Sample JSON request

```
{
  "action": "Suspend",
  "suspend": {
    "metro": true,
```

```
    "dr": true
  }

}
```

2.3.2.4 POST

Resource	metrodr
Request Type:	POST
New Endpoint Description:	Creates a MetroDR Environment: • Ability to create a new MetroDR environment from an unprotected Storage Group • Ability to create a new MetroDR environment from an R11 Storage Group – Storage Group must have an Active leg and an Async or Adaptive Copy leg
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL:	…/92/replication/symmetrix/{symmetrixId}/metrodr
User Roles:	Administrator/Storage Administrator
Prerequisite:	Not applicable
Path Parameters:	Attribute Name | Type | Optional | Description symmetrixId | String | No | Unique identifier of the symmetrix
Query Parameters:	Not applicable
Request Parameters:	Object: MetroDrEnvironmentCreate Name | Type | Optional | Description action | MetroDRCreateAction | No | The action to be performed

create_environment_param	StorageGroupMetroDrEnvironmentCreate	Yes	Details needed to create a new MetroDR Environment
convert_to_metrodr_param	SgConvertMetroDRParam	Yes	Details needed to convert to a MetroDR Environment

Object	MetroDRCreateAction		
Name	Type	Optional	Description
CreateEnvironment	String	Yes	Action for Creating a new MetroDR Environment
ConvertToMetroDR	String	Yes	Action for converting to a MetroDR Environment

Object	StorageGroupMetroDrEnvironmentCreate		
Name	Type	Optional	Description
storage_group_name	String	No	Name that uniquely identifies the source Storage Group.
environment_name	String	No	The unique name of the New MetroDR Environment
metro_r2_array_id	String	No	The 12-digit ID of the Metro R2 Array
dr_array_id	String	No	The 12-digit ID of the DR Array
metro_r1_metro_r2_rdfg_number	Long	Yes	Specify the RDF Group number to use to create the pairs, or leave blank to automatically pick or create an SRDF Group for the setup
force_new_metro_r1_metro_r2_rdfg	Boolean	Yes	Overrides the SRDF Group selection functionality and forces the creation of a new SRDF Group
metro_r1_dr_rdfg_number	Long	Yes	Specify the RDF Group number to

			use to create the pairs, or leave blank to automatically pick or create an SRDF Group for the setup
force_new_metro_r1_dr_rdfg	Boolean	Yes	Overrides the SRDF Group selection functionality and forces the creation of a new SRDF Group
metro_r2_dr_rdfg_number	Long	Yes	Specify the RDF Group number to use to create the pairs, or leave blank to automatically pick or create an SRDF Group for the setup
force_new_metro_r2_dr_rdfg	Boolean	Yes	Overrides the SRDF Group selection functionality and forces the creation of a new SRDF Group
dr_replication_mode	MetroDrRdfMode	Yes	The replication mode the DR SRDF pairs will be created
metro_establish	Boolean	Yes	Specifies if the pairs are to be established on creation
dr_establish	Boolean	Yes	Specifies if the pairs are to be established on creation
metro_r2_storage_group_name	String	No	The name of the storage group to be created on the remote Symmetrix
metro_r2_sl	String	Yes	The name of the SLO on the remote Symmetrix that the storage group will be given
metro_r2_no_compression	Boolean	Yes	Do not use compression on the remote storage group when the remote array SRP is compression capable
dr_storage_group_name	String	No	The name of the storage group to be created on the DR array

dr_sl	String	Yes	The name of the SLO on the remote Symmetrix that the storage group will be given
dr_no_compression	Boolean	Yes	Do not use compression on the remote storage group when the remote array SRP is compression capable

Object	MetroDrRdfMode		
Name	Type	Optional	Description
ADAPTIVE_COPY_DISK	String	Yes	Use Adaptive Copy Disk mode
ASYNCHRONOUS	String	Yes	Use Asynchronous mode

Object	SgConvertMetroDRParam		
Name	Type	Optional	Description
storage_group_name	String	No	Name that uniquely identifies the source Storage Group.
environment_name	String	No	The unique name of the New MetroDR Environment
metro_r2_dr_rdfg	Integer	No	Local SRDF Group Number of the Metro R2 to DR SRDF Group

Response Parameters :

SYNCHRONOUS:

Object	MetroDrEnvironment		
Attribute Name	Type	Optional	Description
< MetroDrEnvironment > is detailed in GET (Object)			

ASYNCHRONOUS:

Object	job		
Attribute Name	Type	Optional	Description
<Job> is detailed in the Appendix.			

2.3.2.4.1 Sample JSON request

```
{
  "action": "CreateEnvironment",
```

```
  "create_environment_param": {
    "dr_storage_group_name": "sg_5",
    "storage_group_name": "sg_1",
    "metro_r2_array_id": "007878788333",
    "dr_array_id": "078223654732",
    "metro_r2_storage_group_name": "sg_5",
    "environment_name": "env_1"
  }
}
```

2.3.2.5 DELETE

Resource	N/A
Request Type	DELETE
New Endpoint Description:	Ability to remove (delete) a MetroDR Environment This opereation wil remove one SRDF group from the MetroDR Environment and leave the remaining groups in place – this will result in Metro with DR on either the R1 or R2 side. Default behavior is to remove the MetroR2 to DR SRDF Group
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL	…/92/replication/symmetrix/{symmetrixId}/metrodr/{environmentName}
User Roles	StorageAdmin
Prerequisite	Not applicable
Path Parameters	No change
Request Parameters	Not applicable

Query Parameters

Name	Type	Optional	Description
remove_r1_dr_rdfg	String	Yes	Removes the MetroR1 to DR SRDF Group from the Environment instead
force	String	Yes	Attempts to force the operation even though one or more volumes may not be in the normal,

				expected state(s) for the specified operation.
	symforce	String	Yes	Forces the operation to execute when normally it is rejected. Use extreme caution with this option. If used when a link is copy in progress or when a restore is restore in progress, this will cause an incomplete copy and data on the copy target would not be usable.
Response Parameters	No content			

2.3.3 snapshot_policy

2.3.3.1 GET (List)

Resource	snapshot_policy
Request Type:	GET
New Endpoint Description:	Get the list of snapshot policies on the array
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL:	…/92/replication/symmetrix/{symmetrixId}/snapshot_policy
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable
Path Parameters:	Attribute Name | Type | Optional | Description

	symmetrixId	String	No	Unique identifier of the symmetrix	
Query Parameters:	Not applicable				
Request Parameters:	Not applicable				
Response Parameters:					

Object	SnapshotPolicyList		
Name	Type	Optional	Description
names	List<String>	No	List of snapshot policy names

2.3.3.1.1 Sample JSON response

```
{
   "names": [ “Name1”, “Name2” ]
}
```

2.3.3.2 GET (Object)

Resource	snapshot_policy
Request Type:	GET
New Endpoint Description:	Get snapshot policy details
Modify Endpoint Description:	

Added in this Release
Not applicable
Modified in this Release
Not applicable
Removed in this Release
Not applicable

URL:	…/92/replication/symmetrix/{symmetrixId}/snapshot_policy/{snapshotPolicyId}
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor

Prerequisite: Not applicable

Path Parameters:

Attribute Name	Type	Optional	Description
symmetrixId	String	No	Unique identifier of the symmetrix
snapshotPolicyId	String	No	Unique policy name

Query Parameters: Not applicable

Request Parameters: Not applicable

Response Parameters:

Object	SnapshotPolicy		
Name	Type	Optional	Description
symmetrixID	String	No	The array that the SP lives on
snapshot_policy_name	String	No	The SP name
snapshot_count	Integer	No	The maximum number of snapshots
interval_minutes	Integer	No	Interval between snapshots
offset_minutes	Integer	No	Minutes in interval the snapshot will be taken
suspended	Boolean	Yes	Is the SP suspended
secure	Boolean	Yes	Does the SP create secure snapshots
last_time_used	String	Yes	The last time the SP was used
storage_group_count	Integer	Yes	The number of SGs that this SP is directly associated with.
compliance_count_warning	Integer	Yes	The count below which

				compliance will become warning.
	compliance_count_critical	Integer	Yes	The count below which compliance will become critical

2.3.3.2.1 Sample JSON response


```
{
    "symmetrixID": "000197900256",
    "snapshot_policy_name": "DailyDefault",
    "snapshot_count": 14,
    "interval_minutes": 1440,
    "offset_minutes": 1381,
    "compliance_count_critical": 10
}
```


2.3.3.3 POST

Resource	snapshot_policy
Request Type:	POST
New Endpoint Description:	Create a snapshot policy
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL:	…/92/replication/symmetrix/{symmetrixId}/snapshot_policy
User Roles:	Administrator/Storage Administrator
Prerequisite:	Not applicable

Path Parameters:

Attribute Name	Type	Description
symmetrixId	String	Unique identifier of the symmetrix

Query Parameters:	Not applicable

Request Parameters:

Object	SnapshotPolicyCreate		
Name	Type	Optional	Description
snapshot_policy_id	string	No	The name of the new SP
cloud_snapshot_policy_details	Cloud_Snapshot_Policy_Details	Yes	Details needed to create a cloud policy
local_snapshot_policy_details	Local_Snapshot_Policy_Details	Yes	Details needed to create a local policy
interval	Enum (SnapshotInterval) [10 Minutes,12 Minutes, 15 Minutes,20 Minutes, 30 Minutes,1 Hour, 2 Hours,3 Hours, 4 Hours, 6 Hours, 8 Hours, 12 Hours, 1 Day, 7 Days]	No	The value of the interval counter for snapshot policy execution
offset_mins	int	No	The number of minutes after 00:00 on Monday to first run the SP.
compliance_count_warning	Int	No	The Number of snapshots which are not failed or bad when compliance changes to warning.
compliance_count_critical	Int	No	The Number of snapshots which are not failed or bad when compliance

			changes to critical.

Object	Cloud_Snapshot_Policy_Details		
Name	Type	Optional	Description
cloud_retention_days	Integer	No	The number of days to retain the snapshot
cloud_provider_name	String	No	Name of the cloud provider

Object	Local_Snapshot_Policy_Details		
Name	Type	Optional	Description
secure	Boolean	Yes	Create secure snapshots
snapshot_count	Integer	No	Number of snapshots to retain

Response Parameters:

SYNCHRONOUS:

Object	SnapshotPolicy		
Attribute Name	Type	Optional	Description
(see GET (Object) for object details)			

ASYNCHRONOUS:

Object	job		
Attribute Name	Type	Optional	Description
<Job> is detailed in the Appendix.			

2.3.3.3.1 Sample JSON request

(see GET (Object) for object details)

2.3.3.4 PUT

Resource	snapshot_policy
Request Type:	PUT

<table>
<tr><td>New Endpoint Description:</td><td>Modify a Snapshot Policy by:
changing its attributes (modify)
suspend or resume the SP
associate the SP with Storage Groups (addSGs)
dis-assocaited the SP from Storage Groups (removeSGs).</td></tr>
<tr><td>Modify Endpoint Description:</td><td><table><tr><th>Added in this Release</th></tr><tr><td>Not applicable</td></tr><tr><th>Modified in this Release</th></tr><tr><td>Not applicable</td></tr><tr><th>Removed in this Release</th></tr><tr><td>Not applicable</td></tr></table></td></tr>
<tr><td>URL:</td><td>…/92/replication/symmetrix/{symmetrixId}/snapshot_policy</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td><table><tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr><tr><td>symmetrixId</td><td>String</td><td>No</td><td>Unique identifier of the symmetrix</td></tr><tr><td>snapshotPolicyId</td><td>String</td><td>No</td><td>The policy ID</td></tr></table></td></tr>
<tr><td>Query Parameters:</td><td>Not applicable</td></tr>
</table>

Request Parameters:

Object	SnapshotPolicyUpdate		
Name	Type	Optional	Description
action	Enum (SnapshotPolicyAction) [Modify, Suspend, Resume, AssociateToStorageGroups, DisassociateFromStorageGroups]	No	Action to perform
modify	SnapshotPolicyModify	Yes	Parameters specific to modifying the snapshot policy
addSGs	SnapshotPolicyStorageGroupAddRemove	Yes	List of SG names to associate this SP with.
removeSGs	SnapshotPolicyStorageGroupAddRemove	Yes	List of SG names to dis-associate this SP from.

Object	SnapshotPolicyModify		
Name	Type	Optional	Description
snapshot_policy_name	String	No	Snapshot policy to be modified
interval_mins	Long	Yes	Snapshot policy interval
offset_mins	Integer	Yes	Snapshot policy offset
snapshot_count	Integer	Yes	Maximum number of snapshots to retain
compliance_count_warning	Long	Yes	Snapshot count below which compliance will become warning
compliance_count_critical	Long	Yes	Snapshot count below which compliance will become critical

Object	SnapshotPolicyStorageGroupAddRemove		
Name	Type	Optional	Description
storage-group_names	List<String>	No	List of storage group names

Response Parameters:

SYNCHRONOUS:

Object	SnapshotPolicy

	Attribute Name	Type	Optional	Description
	< SnapshotPolicy> is detailed in GET (Object)			

ASYNCHRONOUS:

Object	job		
Attribute Name	Type	Optional	Description
<Job> is detailed in the Appendix.			

2.3.3.4.1 Sample JSON request

```
{
  "associate_to_storage_group": {
    "storage_group_name": [
      "sg_1"
    ]
  },
  "action": "AssociateToStorageGroups"

}
```

2.3.3.5 DELETE

Resource	snapshot_policy
Request Type:	DELETE
New Endpoint Description:	Delete a snapshot policy
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL:	…/92/replication/symmetrix/{symmetrixId}/snapshot_policy/{snapshotPolicyId}
User Roles:	Administrator/Storage Administrator
Prerequisite:	Not applicable
Path Parameters:	

	Attribute Name	Type	Optional	Description
	symmetrixId	String	No	Unique identifier of the symmetrix
	snapshotPolicyId	String	No	Unique identifier of the Snapshot Policy
Query Parameters:	Not applicable			
Request Parameters:	Not applicable			
Response Parameters:	No content			

2.3.3.6 storagegroup

2.3.3.6.1 GET (List)

Resource	storagegroup
Request Type:	GET
New Endpoint Description:	Get the storage group names that the Policy is associated with
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL:	…/92/replication/symmetrix/{symmetrixId}/snapshot_policy/{snapshotPolicyId}/storagegroup
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable

Path Parameters:

Attribute Name	Type	Optional	Description
symmetrixId	String	No	Unique identifier of the symmetrix

	snapshotPolicyId	String	No	Unique identifier of the Snapshot Policy	
Query Parameters:	Not applicable				
Request Parameters:	Not applicable				
Response Parameters:	Not applicable				

2.3.3.6.1.1 Sample JSON Response

```
{
   "name": [“sg_1”]
}
```

2.3.3.6.2 GET (Object)

Resource	snapshot_policy
Request Type:	GET
New Endpoint Description:	Get snapshot policy storage group details
Modify Endpoint Description:	Added in this Release Not Applicable Modified in this Release Not Applicable Removed in this Release Not Applicable
URL:	…/92/replication/symmetrix/{symmetrixId}/snapshot_policy/{snapshotPolicyId}/storagegroup/{storageGroupId}
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable

Path Parameters:

Attribute Name	Type	Optional	Description
symmetrixId	String	No	Unique identifier of the symmetrix
snapshotPolicyId	String	No	Unique identifier of the Snapshot Policy
storageGroupId	String	No	Unique identifier of the Storage Group

Query Parameters: Not applicable

Request Parameters: Not applicable

Response Parameters:

Object	SnapshotPolicyStorageGroup		
Name	Type	Optional	Description
symmetrixID	String	No	The array that the SP lives on
snapshot_policy_id	String	No	The SP name
storage_group_id	String	No	The SG name
suspended	Boolean	Yes	Is the SP suspended
compliance	Compliance	Yes	The compliance level for this SG/SP combination.
snapshots_in_time_window	Integer	Yes	Number of snapshots in the time window
error_messages	List<String>	Yes	Compliance error messages

2.3.3.6.2.1 Sample JSON Response

```
{
   "symmetrixID": "000197900256",
   "snapshot_policy_id": "SmallCounts",
   "storage_group_id": "BOH_SVX_SG",
   "total_snapshots": 5
}
```

2.3.4 rdf_group

2.3.4.1 DELETE

Resource	rdf_group
Request Type:	DELETE

New Endpoint Description:	Not applicable
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release force
URL:	…/92/replication/symmetrix/{symmetrixId}/rdf_group/{rdfgNumId}
User Roles:	StorageAdmin
Prerequisite:	Not applicable
Path Parameters:	Not applicable
Query Parameters:	Not applicable
Request Parameters:	Not applicable
Response Parameters:	Not applicable

2.3.4.2 GET

Resource	rdf_group
Request Type:	GET
New Endpoint Description:	Not applicable
Modify Endpoint Description:	Added in this Release vasa_group link_limbo hardware_compression software_compression

	offline
	Modified in this Release
	Not applicable
	Removed in this Release
	Not applicable
URL:	…/92/replication/symmetrix/{symmetrixId}/rdf_group/{rdfgNumId}
User Roles:	Monitor
Prerequisite:	Not applicable
Query Parameters:	Not applicable
Request Parameters:	Not applicable
Response Parameters:	

Object	RdfGroup		
Attribute Name	Type	Optional	Description
vasa_group	boolean	Yes	Indicates that the group is a VASA replication group.
link_limbo	long	Yes	Indicates the Link Limbo of the SRDF Group
hardware_compression	boolean	Yes	Is Hardware Compression enabled for the SRDF Group
software_compression	boolean	Yes	Is Software Compression enabled for the SRDF Group
offline	boolean	Yes	Is the SRDF group online or offline
rdf_properties	RdfGroupAsync	Yes	Properties of Asynchronous Rdf Group

Object	RdfGroupAsync		
Attribute Name	Type	Optional	Description
session_uncommitted_tracks	Long	Yes	Session uncommited tracks

	r1_to_r2_lag_time	Long	Yes	R1 to R2 Lag Time
	duration_of_last_cycle	Long	Yes	Duration of last cycle
	average_cycle_time	Long	Yes	Average cycle time
	duration_of_last_transmit_cycle	Long	Yes	Duration of last transmit cycle
	transmit_queue_depth	Long	Yes	Transmit queue depth
	cycle_number	Long	Yes	Cycle number
	transmit_idle_state	String	Yes	Transmit idle state
	transmit_idle_time	Long	Yes	Transmit idle time
	consistency_protection	String	Yes	Consistency protection
	dse_active	Boolean	Yes	Is dse Active
	dse_threshold	Long	Yes	Dse threshold
	dse_autostart	String	Yes	Dse autostart
	session_priority	Long	Yes	Session priority
	consistency_exempt_volumes	Boolean	Yes	Has consistency exempt volumes

2.3.4.2.1 Sample JSON Response


```
"vasa_group": false,
"link_limbo": 10,
"hardware_compression": false,
"software_compression": false,
"offline": false
"rdfa_properties": {
        "session_uncommitted_tracks": 0,
        "r1_to_r2_lag_time": 28,
        "duration_of_last_cycle": 15,
        "average_cycle_time": 15,
        "duration_of_last_transmit_cycle": 15,
        "transmit_queue_depth": 1,
        "cycle_number": 90,
        "transmit_idle_state": "Enabled",
        "transmit_idle_time": 0,
        "consistency_protection": "Enabled",
```

```
    "dse_active": true,
    "dse_threshold": 50,
    "dse_autostart": "Enabled",
    "session_priority": 33,
    "consistency_exempt_volumes": false
}
```

2.3.4.3 PUT

Resource	RdfGroupUpdate
Request Type	PUT
New Endpoint Description:	Not applicable
Modify Endpoint Description:	Added in this Release RdfGroupAction.SetAttributes RdfGroupAction.EnableDse set_attributes enable_dse Modified in this Release Not applicable Removed in this Release Not applicable
URL	…/92/replication/symmetrix/{symmetrixId}/rdf_group/{rdfgNumId}
User Roles	StorageAdmin
Prerequisite	Not applicable

Path Parameters

Attribute Name	Type	Description
symmetrixId	String	Unique identifier of the symmetrix
rdfgNumId	String	The SRDF Group Number

Query Parameters: Not applicable

Request Parameters

Object	RdfGroupUpdate		
Name	Type	Optional	Description
action	RdfGroupAction	No	The action to be performed.

set_attributes	RdfGroupSetAttributesParam	Yes	Set Attributes on the SRDF group
enable_dse	RdfGroupEnableDSEParam	Yes	Enable/ Disable DSE on the SRDF group

Object	RdfGroupAction		
Name	Type	Optional	Description
SetAttributes	String	Yes	Action for setting attributes on the SRDF group
EnableDse	String	Yes	Action for SRDF/A DSE adjusting on the group

Object	RdfGroupSetAttributesParam		
Name	Type	Optional	Description
link_limbo	long	Yes	Sets the link limbo for the SRDF Group
hardware_compression	boolean	Yes	Enable/ Disable Hardware Compression for the SRDF Group
software_compression	boolean	Yes	Enable/ Disable Software Compression for the SRDF Group
dse_threshold	int	Yes	Sets the DSE Threshold for the SRDF Group
dse_autostart	boolean	Yes	Enable/ Disable DSE Autostart for the SRDF Group
session_priority	int	Yes	Set the SRDF/A Session Priority for the SRDF Group
sym_force	boolean	Yes	Requests the Symmetrix array force operation be executed when normally it is rejected. Use extreme caution when using option. CAUTION: Use care when applying symforce, as data could be lost or corrupted. Use of this option is not recommended, except in an emergency. NOTE: To enable symforce, a parameter called SYMAPI_ALLOW_RDF_SYMFORCE in the options file must be set to TRUE and restart the Unisphere server to pick up the change.

	Object	RdfGroupEnableDSEParam		
	Name	Type	Optional	Description
	enable	boolean	No	Enables/ Disables DSE for the SRDF Group
Response Parameters:	RdfGroup			

2.3.4.3.1 Sample JSON Request (Set Attributes)

```
{
   "action": "SetAttributes",
   "set_attributes":
   {
      "link_limbo": 10,
      "hardware_compression": false,
      "software_compression": false,
      "dse_threshold" : 10,
      "dse_autostart" : false,
      "session_priority" : 10,
      "sym_force" : false
   }
}
```

2.3.4.3.2 Sample JSON Request (Enable DSE)

```
{
   "action": "EnableDse",
   "enable_dse":
   {
      "enable": true
   }
}
```

2.4 Migration

2.5 vVols

2.5.1 replication_group

2.5.1.1 GET (List)

Resource	replication_group

<table>
<tr><td>Request Type:</td><td>GET</td></tr>
<tr><td>New Endpoint Description:</td><td>Gets a list of VASA Replication group names</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>…/92/vvol/symmetrix/{symmetrixId}/replication_group</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>Unique identifier of the symmetrix</td></tr>
</table>
</td></tr>
<tr><td>Query Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr>
<tr><td>Remote_symmetrix_id</td><td>String</td><td>An optional value that filters the returned list so it displays VASA Replication Groups where the remote symm of the RDF Group matches ‘remote_symmetrix_id’</td></tr>
</table>
</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>
<table>
<tr><th>Object</th><td colspan="3">RepGroupLabelList</td></tr>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>rep_group_id</td><td>ReplicationGroupID array</td><td>No</td><td>A list of VASA Replication Group ID Objects</td></tr>
<tr><td colspan="4"></td></tr>
<tr><th>Object</th><td colspan="3">ReplicationGroupID</td></tr>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>rep_grp_number</td><td>int</td><td>No</td><td>The replication group number</td></tr>
</table>
</td></tr>
</table>

	label	string	No	The replication group label

Sample JSON response


```
{
   "rep_group_id": [
      {
         "rep_grp_number": 7,
         "label": "test"
      },
      {
         "rep_grp_number": 8,
         "label": "Ctest"
      },
      {
         "rep_grp_number": 13,
         "label": "DaveH_RepG"
      },
      {
         "rep_grp_number": 14,
         "label": "DH_RefGrp"
      },
      {
         "rep_grp_number": 17,
         "label": "new"
      },
      {
         "rep_grp_number": 46,
         "label": "Multiple3"
      }
   ]
}
```


2.5.1.2 GET (Object)

Resource	replication_group
Request Type:	GET

<table>
<tr><td>New Endpoint Description:</td><td>Gets the details of a single replication group</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>…/92/vvol/symmetrix/{symmetrixId}/replication_group/{repGrpNumId}</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>Not applicable</td></tr>
<tr><td>Query Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr>
<tr><td>remote_symmetrix_id</td><td>String</td><td>An optional value that filters the returned list so it displays VASA Replication Groups where the remote symm of the RDF Group matches ‘remote_symmetrix_id’</td></tr>
</table>
</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>
<table>
<tr><th>Object</th><td colspan="3">ReplicationGroup</td></tr>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>label</td><td>string</td><td>No</td><td>The label that the Replication group will have. Must be unique to both local and remote arrays.</td></tr>
<tr><td>local_rep_group_number</td><td>int</td><td>No</td><td>The Replication group number for the local array</td></tr>
<tr><td>local_storage_container</td><td>string</td><td>No</td><td>The Local Storage Container used to create the Vasa Replication Group</td></tr>
<tr><td>local_ports</td><td>String Array</td><td>No</td><td>A collection of SRDF ports to be</td></tr>
</table>
</td></tr>
</table>

				used on the local side
	local_online_ports	String Array	No	A collection of online SRDF ports to be used on the local side
	remote_rep_group_number	int	No	The Replication group number for the remote array
	remote_storage_container	string	No	The Remote Storage Container used to create the Vasa Replication Group
	remote_symmetrix_id	string	No	Remote Symmetrix ID
	remote_ports	String Array	No	A collection of SRDF ports to be used on the remote side
	remote_online_ports	String Array	No	A collection of online SRDF ports to be used on the remote side

Sample JSON response


```
{
    "label": "test",
    "local_rep_group_number": 7,
    "local_storage_container": "N/A",
    "remote_rep_grp_number": 1,
    "remote_symmetrix": "000197900256",
    "remote_storage_container": "N/A",
    "local_ports": [
        "RF-1E:7"
    ],
    "remote_ports": [],
    "local_online_ports": [
        "RF-1E:7"
    ],
    "remote_online_ports": []
}
```

2.5.1.1 POST

<table>
<tr><td>Resource</td><td>replication_group</td></tr>
<tr><td>Request Type</td><td>POST</td></tr>
<tr><td>New Endpoint Description:</td><td>Creates a single of VASA Replication group</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL</td><td>…/92/vvol/symmetrix/{symmetrixId}/replication_group</td></tr>
<tr><td>User Roles</td><td>StorageAdmin</td></tr>
<tr><td>Prerequisite</td><td>Not applicable</td></tr>
<tr><td>Path Parameters</td><td>symmetrixId</td></tr>
<tr><td>Request Parameters</td><td>
<table>
<tr><th>Object</th><th colspan="3">ReplicationGroupCreate</th></tr>
<tr><th>Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>label</td><td>string</td><td>No</td><td>The label that the Replication group will have. Must be unique to both local and remote arrays.</td></tr>
<tr><td>local_rep_group_number</td><td>int</td><td>No</td><td>The Replication group number for the local array</td></tr>
<tr><td>local_storage_container</td><td>string</td><td>No</td><td>The Local Storage Container used to create the Vasa Replication Group</td></tr>
<tr><td>local_ports</td><td>RepGrpDirectorPort</td><td>No</td><td>A collection of SRDF ports to be used on the local side. All ports must be from the same array and must be the same configuration wrt Fibre/GigE</td></tr>
<tr><td>remote_rep_group_number</td><td>int</td><td>No</td><td>The Replication group number for the remote array</td></tr>
<tr><td>remote_storage_container</td><td>string</td><td>No</td><td>The Remote Storage Container used to create the Vasa Replication Group</td></tr>
<tr><td>remote_symmetrix_id</td><td>string</td><td>No</td><td>Remote Symmetrix ID</td></tr>
</table>
</td></tr>
</table>

	remote_ports	RepGrpDirectorPort	No	A collection of SRDF ports to be used on the local side. All ports must be from the same array and must be the same configuration wrt Fibre/GigE
	local_link_limbo	int	Yes	Link Limbo for the local Replication Group
	remote_link_limbo	int	Yes	Link Limbo for the remote Replication Group
	sw_compression_enabled	Boolean	Yes	Enable Software Compression
	hw_compression_enabled	Boolean	Yes	Enable Hardware Compression
	Object	RepGrpDirectorPort		
	Name	Type	Optional	Description
	director_id	string	No	The identifier for the SRDF Director
	port_number	int	No	The SRDF Director port number.
Query Parameters	Not applicable			
Response Parameters	RdfGroup			

2.5.1.1.1 Sample JSON response

```
{
  "label": "test4",
  "local_rep_group_number": 90,
  "local_storage_container": "testSC",
  "local_ports": [
    {
      "director_id": "RF-2E",
      "port_number": 7,
    }
  ],
  "remote_rep_group_number": 91,
  "remote_storage_container": "Test",
  "remote_symmetrix_id": "000197900049",
  "remote_ports": [
    {
      "director_id": "RF-1E",
      "online": true,
    }
  ]
}
```

2.5.1.2 PUT

<table>
<tr><td>Resource</td><td>replication_group</td></tr>
<tr><td>Request Type</td><td>PUT</td></tr>
<tr><td>New Endpoint Description:</td><td>Ports can be added or removed from a Vasa Replication Group</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL</td><td>…/91/vvol/symmetrix/{symmetrixId}/replication_group/{repGrpNumId}</td></tr>
<tr><td>User Roles</td><td>StorageAdmin</td></tr>
<tr><td>Prerequisite</td><td>Not applicable</td></tr>
<tr><td>Path Parameters</td><td>No change</td></tr>
<tr><td>Request Parameters</td><td>
<table>
<tr><th>Object</th><th colspan="3">ReplicationGroupUpdate</th></tr>
<tr><th>Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>action</td><td>ReplicationGroup Action</td><td>No</td><td>Add ports option</td></tr>
<tr><td>addPorts</td><td>RepGroupAddPor tsParam</td><td></td><td>• Collectionof RepGrpDirectorPorts
• Star option</td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><th>Object</th><th colspan="3">ReplicationGroupAction</th></tr>
<tr><th>Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>action</td><td>string</td><td>No</td><td>‘AddPorts’ or ‘RemovePorts’</td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><th>Object</th><th colspan="3">RepGroupAddPortsParam</th></tr>
<tr><th>Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>ports</td><td>RepGrpDirectorP ort</td><td>No</td><td>SRDF ports to be added to the Replication Group</td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><th>Object</th><th colspan="3">RepGrpDirectorPort</th></tr>
<tr><th>Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>director_id</td><td>string</td><td>No</td><td>The identifier for the SRDF Director</td></tr>
</table>
</td></tr>
</table>

	port_number	int	No	The SRDF Director port number.
Query Parameters	Not applicable			
Response Parameters	RdfGroup			

2.5.1.2.1 Sample JSON Request (Add)

```
{
  "action": "AddPorts",
  "add_ports": {
    "ports": [
      {
        "director_id": "RF-2E",
        "online": true,
      }
    ]
  }
}
```

2.5.1.2.2 Sample JSON Request (Remove)

```
{
  "action": "RemovePorts",
  "remove_ports": {
    "ports": [
      {
        "director_id": "RF-2E",
        "online": true,
       }
    ]
  }

}
```

2.5.1.1 DELETE

Resource	replication_group
Request Type	DELETE
New Endpoint Description:	Delete a VASA Replication Group

Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL	…/92/vvol/symmetrix/{symmetrixId}/replication_group/{repGrpNumId}
User Roles	StorageAdmin
Prerequisite	Not applicable
Path Parameters	No change
Request Parameters	Not applicable
Query Parameters	symforce
Response Parameters	Not applicable

2.5.2 vasaprovider

2.5.2.1 GET (Object)

Resource	vasaprovider
Request Type:	GET
New Endpoint Description:	Not applicable
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release This now brings back an array of up to two objects where before it brought back a list of attributes

<table>
<tr><td></td><td><table><tr><th>Removed in this Release</th></tr><tr><td>Not applicable</td></tr></table></td></tr>
<tr><td>URL:</td><td>…/92/vvol/symmetrix/{symmetrixId}/vasaprovider</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td><table><tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr><tr><td></td><td></td><td></td></tr></table></td></tr>
<tr><td>Query Parameters:</td><td>N/A</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td><table><tr><th>Object</th><th colspan="3">ListVasaProviderResult</th></tr><tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr><tr><td>ipaddress</td><td>String</td><td>No</td><td>The IP Address of the VASA Provider</td></tr><tr><td>status</td><td>String</td><td>No</td><td>The Online/Offline status of the VASA Provider. Can be “ONLINE”, “OFFLINE” or “UNKNOWN”</td></tr><tr><td>availabilityStatus</td><td>String</td><td>Yes</td><td>The availability status of an Embedded VASA provider.</td></tr></table></td></tr>
</table>

2.5.2.1.1 Sample JSON Response (Embedded VASA Provider)

```
{
   "vasaProviders": [
      {
         "ipaddress": "10.60.140.10",
         "status": "ONLINE",
         "availabilityStatus": "Active"
      },
      {
         "ipaddress": "10.60.140.10",
         "status": "ONLINE",
         "availabilityStatus": "Suspended"
      }
   ]
}
```

2.5.2.1.2 Sample JSON Response (External VASA Provider)

```
{
   "vasaProviders": [
      {
         "ipaddress": "10.60.141.145",
```

```
            "status": "UNKNOWN"
        }
    ]
}
```


2.6 System

2.6.1 cloud

2.6.1.1 GET (Object)

Resource	cloud
Request Type:	GET
New Endpoint Description:	Get Cloud System Details
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL:	…/92/system/symmetrix/{symmetrixId}/cloud
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable

Path Parameters:

Attribute Name	Type	Optional	Description
symmetrixId	String	No	Unique identifier of the symmetrix

Query Parameters:	Not applicable
Request Parameters:	Not applicable

Response Parameters:

Object	cloudSystem		
Attribute Name	Type	Optional	Description

cloud_state	String [SETUP_REQUIRED ONLINE, RESOLVABLE, NETWORK_RESOLVABLE ERROR]	No	The current State of the Cloud Environment
info	String	No	A description of the The Cloud System State
num_of_cloud_storage_groups	long	Yes	The total number of Cloud Storage Group that have cloud snapshots
num_of_cloud_snapshots	long	Yes	The total number of Cloud cloud snapshots
num_of_cloud_providers	long	Yes	The total number of Cloud Providers

2.6.1.1.1 Sample JSON response

```
{
        "cloud_state":"ONLINE",
       “info”: “Cloud System is online.”
        "num_of_cloud_storage_groups":3,
        "num_of_cloud_snapshots":10,
        "num_of_cloud_providers":1
}
```

2.6.1.2 POST

<table>
<tr><td>Resource</td><td>cloud</td></tr>
<tr><td>Request Type:</td><td>POST</td></tr>
<tr><td>New Endpoint Description:</td><td>Creates a Cloud Environment on the Array and Adds a Cloud System to the Unisphere instance</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>… /92/system/symmetrix/{symmetrixId}/cloud</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>Unique identifier of the symmetrix</td></tr>
</table>
</td></tr>
<tr><td>Query Parameters:</td><td>Not applicable</td></tr>
<tr><td>Request Parameters:</td><td>
<table>
<tr><th>Object</th><th colspan="3">CreateCloudSystemParam</th></tr>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>setup_cloud_system_interfaces</td><td>SetupCloudSystemInterfaceParamType</td><td>YES</td><td>Optional Param to Sets the IP Address of the Cloud System interfaces</td></tr>
</table>
</td></tr>
</table>

create_cloud_system_dns_server	CreateCloudSystemDnsServerParamType	YES	Optional Param to add a DNS Server to the Cloud System
create_cloud_system_route	CreateCloudSystemRouteParamType	YES	Optional Param to add a IP Route to the Cloud System
edit_bandwidth_limit	EditBandwidthLimitParamType	YES	Optional Param to Set Bandwidth Limits on the Cloud System

Object	SetupCloudSystemInterfaceParamType		
Attribute Name	Type	Optional	Description
cloud_system_interface_ip_settings	Array [CloudSystemInterfaceIpSettingsParamType]	NO	List of interface settings

Object	CloudSystemInterfaceIpSettingsParamType		
Attribute Name	Type	Optional	Description
interface_id	CloudSystemInterfaceIdType	NO	The interface id to be used
edit_interface_ip_settings	EditInterfaceIpSettingsParamType	NO	The setting to be applied to the interface

	<table><tr><td>Object</td><td colspan="3">CreateCloudSystemRouteParamType</td></tr><tr><td>Attribute Name</td><td>Type</td><td>Optional</td><td>Description</td></tr><tr><td>create_cloud_system_dns_server</td><td>String</td><td>The IP Address of the DNS Server</td><td>The IP Address of the DNS Server</td></tr></table>
Response Parameters:	SYNCHRONOUS: <table><tr><td>Object</td><td colspan="3">cloudSystem</td></tr><tr><td>Attribute Name</td><td>Type</td><td>Optional</td><td>Description</td></tr><tr><td colspan="4">(see GET (Object) for object details)</td></tr></table> ASYNCHRONOUS: <table><tr><td>Object</td><td colspan="3">job</td></tr><tr><td>Attribute Name</td><td>Type</td><td>Optional</td><td>Description</td></tr><tr><td colspan="4"><Job> is detailed in the Appendix.</td></tr></table>

2.6.1.2.1 Sample JSON request

TODO

2.6.1.3 PUT

Resource	cloud
Request Type:	PUT
New Endpoint Description:	Modifies the Cloud System
Modify Endpoint Description:	<table><tr><td>Added in this Release</td></tr><tr><td>Not applicable</td></tr><tr><td>Modified in this Release</td></tr></table>

	Not applicable
	Removed in this Release
	Not applicable
URL:	…/92/system/symmetrix/{symmetrixId}/cloud/
User Roles:	Administrator/Storage Administrator
Prerequisite:	Not applicable

Path Parameters:

Attribute Name	Type	Optional	Description
symmetrixId	String	No	Unique identifier of the symmetrix

Query Parameters: Not applicable

Request Parameters:

Object	EditCloudSystemParam		
Attribute Name	**Type**	**Optional**	**Description**
action	EditCloudSystemActionParamType	No	Select one of the Edit Options to modify the specified Cloud System

Object	EditCloudSystemActionParamType		
Attribute Name	**Type**	**Optional**	**Description**
resolve	String	NO	Resolves any issues that can be resolved when the Cloud System is in a RESOVABLE state
rerun_setup	String	NO	Reruns the Cloud System setup to re-establish communication between U4P and the Cloud

				System, this should only be executed if the the enrollment has been reset on the cloud system by DELL EMC support

Response Parameters:

SYNCHRONOUS:

Object	CloudSystem		
Attribute Name	Type	Optional	Description
<CloudSystem> is detailed in GET (Object)			

ASYNCHRONOUS:

Object	job		
Attribute Name	Type	Optional	Description
<Job> is detailed in the Appendix.			

2.6.1.3.1 Sample JSON request

```
{
  "action": "resolve”
}
```

2.6.1.4 DELETE

Resource	cloudprovider
Request Type:	DELETE
New Endpoint Description:	Remove Cloud System from Unisphere instance. Note: All Cloud related objects such as Cloud Policies, Cloud Providers and Cloud snapshots must be removed from the array before executing this command
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release

<table>
<tr><td></td><td>Not applicable
Removed in this Release
Not applicable</td></tr>
<tr><td>URL:</td><td>… /92/system/symmetrix/{symmetrixId}/cloud</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td><table><tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr><tr><td>symmetrixId</td><td>String</td><td>No</td><td>Unique identifier of the symmetrix</td></tr></table></td></tr>
<tr><td>Query Parameters:</td><td>Not applicable</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>No content</td></tr>
</table>

2.6.2 cloudjob

2.6.2.1 GET (List)

<table>
<tr><td>Resource</td><td>cloudjob</td></tr>
<tr><td>Request Type:</td><td>GET</td></tr>
<tr><td>New Endpoint Description:</td><td>Gets a List of Cloud Sytem Jobs</td></tr>
<tr><td>Modify Endpoint Description:</td><td><table><tr><th>Added in this Release</th></tr><tr><td>Not applicable</td></tr><tr><th>Modified in this Release</th></tr><tr><td>Not applicable</td></tr><tr><th>Removed in this Release</th></tr><tr><td>Not applicable</td></tr></table></td></tr>
<tr><td>URL:</td><td>…/92/system/symmetrix/{symmetrixId}/cloud/cloudjob</td></tr>
</table>

User Roles: Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor

Prerequisite: Not applicable

Path Parameters:

Attribute Name	Type	Optional	Description
symmetrixId	String	No	Unique identifier of the symmetrix

Query Parameters:

Attribute Name	Type	Description
cloud_snapshot_name	String	Optional value that filters returned list to display Cloud Job infos that have a specified cloud_snapshot_name example: equal to "cloud_snapshot_name=monday_hourly_backup" or contains "cloud_snapshot_name=<like>monday"
state	String	Optional value that filters returned list to display Cloud Job infos that have a specified State - example: equal to "state=running" or contains "state=<like>run"
type	String	Optional value that filters returned list to display Cloud Job infos that have a specified Type example equal to "type=protect" or contains "state=<like>pro"
cloud_snapshot_id	String	Optional value that filters returned list to display Cloud Job infos that have a specified cloud_snapshot_id example equal to "cloud_snapshot_id=105573710593" or contains "cloud_snapshot_id=<like>123"
cloud_provider_id	String	Optional value that filters returned list to display Cloud Job infos that have a specified cloud_provider_id example equal to "cloud_provider_id=ecs1234" or contains "cloud_provider_id=<like>ecs"
cloud_provider_uuid	String	Optional value that filters returned list to display Cloud Job infos that have a specified cloud_provider_uuid example equal to "cloud_provider_uuid=479cb4f4-a6db-11e9-ae21-005056a64358" or contains "cloud_provider_uuid=<like>479cb4f4"
storage_group_id	String	Optional value that filters returned list to display Cloud Job infos that have a specified storage_group_name

		example equal to "storage_group_name=Sales_Production_1" or contains "storage_group_name=<like>Sales"
storage_group_uuid	String	Optional value that filters returned list to display Cloud Job infos that have a specified storage_group_uuid example equal to "storage_group_uuid=60000970000197900256533030323237" or contains "storage_group_uuid=<like>123"
start_date_milliseconds	Long	Optional value that filters returned list to display Cloud Job infos that contain greater than("start_date_milliseconds=>1"), Less than("start_date_milliseconds=<1") or equal to the start_date_milliseconds
completed_date_milliseconds	Long	Optional value that filters returned list to display Cloud Job infos that contain greater than("completed_date_milliseconds=>1"), Less than("completed_date_milliseconds=<1") or equal to the completed_date_milliseconds
num_of_runs	Long	Optional value that filters returned list to display Cloud Job infos that contain greater than("num_of_runs=>1"), Less than("num_of_runs=<1") or equal to the num_of_runs

Request Parameters:

Not applicable

Response Parameters:

Object	CloudJobInfoList		
Attribute Name	Type	Optional	Description
cloud_job_info	Array<CloudJobInfo>	No	A list of Cloud Job Infos

Object	CloudJobInfo		
Attribute Name	Type	Optional	Description
cloud_job_id	String	No	The ID of the Cloud Job
cloud_snapshot_id	String	YES	The id of the Cloud Snapshot

start_date_milliseconds	Long	YES	The Cloud Job Start Date in milliseconds from Epoch
start_date	String	YES	The Cloud Job Start Date
completed_date_milliseconds	Long	YES	The Cloud Job Completed Date in milliseconds from Epoch
completed_date	String	YES	The Cloud Job Completed Date

2.6.2.1.1 Sample JSON response


```
{
   "cloud_job_info": [
      {
         "cloud_job_id": "b26f5847-90ec-4609-b887-f0e06cc6b105",
         "cloud_snapshot_id": "108866771200",
         "start_date_milliseconds": 1569847353000,
         "start_date": "13:42:33 Mon, 30 Sep 2019 IST +0100"
      },
      {
         "cloud_job_id": "e98c467e-090e-49a0-950a-e00b1acc8f46",
         "cloud_snapshot_id": "108822531073",
         "start_date_milliseconds": 1569847238000,
         "start_date": "13:40:38 Mon, 30 Sep 2019 IST +0100"
      }
   ]
}
```


2.6.2.2 GET (Object)

Resource	cloudjob
Request Type:	GET
New Endpoint Description:	Gets the Cloud Job Details
Modify Endpoint Description:	Added in this Release Not applicable

	Modified in this Release
	Not applicable
	Removed in this Release
	Not applicable
URL:	…/92/system/symmetrix/{symmetrixId}/cloud/cloudjob/{cloud_job_id}
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable

Path Parameters:

Attribute Name	Type	Optional	Description
symmetrixId	String	No	Unique identifier of the symmetrix
cloud_job_id	String	No	The ID of the Cloud Job

Query Parameters: Not applicable

Request Parameters: Not applicable

Response Parameters:

Object	CloudJob		
Attribute Name	Type	Optional	Description
cloud_job_id	String	No	The ID of the Cloud Job
cloud_snapshot_name	String	YES	The name of the Cloud Snapshot
cloud_snapshot_id	String	YES	The id of the Cloud Snapshot
cloud_provider_id	String	YES	The id of the Cloud Provider
cloud_provider_uuid	String	YES	The UUID of the Cloud Provider
storage_group_id	String	YES	The id of the Storage Group

storage_group_uuid	String	YES	The UUID of the Storage Group
state	String	YES	The State of the Cloud Job
type	String	YES	The Cloud Job Type
start_date_milliseconds	Long	YES	The Cloud Job Start Date in milliseconds from Epoch
start_date	String	YES	The Cloud Job Start Date
completed_date_milliseconds	Long	YES	The Cloud Job Completed Date in milliseconds from Epoch
completed_date	String	YES	The Cloud Job Completed Date
num_of_runs	Long	YES	The Number of times the Cloud Job has been executed/ retried

2.6.2.2.1 Sample JSON response


```
{
   "cloud_job_id": "b26f5847-90ec-4609-b887-f0e06cc6b105",
   "cloud_snapshot_name": "CloudSnapPolicy",
   "cloud_snapshot_id": "108866771200",
   "cloud_provider_id": "minioecs3",
   "cloud_provider_uuid": "05a014ca-da15-11e9-86a2-49850db0977b",
   "storage_group_id": "shiv_spp_test_1",
   "storage_group_uuid": "600009700bcbb711361101710000000d",
   "type": "Protect",
   "state": "Failed",
   "start_date_milliseconds": 1569847353000,
   "start_date": "13:42:33 Mon, 30 Sep 2019 IST +0100",
   "num_of_runs": 1
}
```

2.6.2.3 PUT

<table>
<tr><td>Resource</td><td>cloudjob</td></tr>
<tr><td>Request Type:</td><td>PUT</td></tr>
<tr><td>New Endpoint Description:</td><td>Modifies the Cloud Job</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>…/92/system/symmetrix/{symmetrixId}/cloud/cloudjob/{cloud_job_id}</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>No</td><td>Unique identifier of the symmetrix</td></tr>
<tr><td>cloud_job_id</td><td>String</td><td>No</td><td>The ID of the Cloud Job</td></tr>
</table>
</td></tr>
<tr><td>Query Parameters:</td><td>Not applicable</td></tr>
<tr><td>Request Parameters:</td><td>
<table>
<tr><th>Object</th><th colspan="3">EditCloudJobParam</th></tr>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>action</td><td>EditCloudJobActionParamType</td><td>No</td><td>Select one of the Edit Options to modify the specified Cloud Job</td></tr>
</table>
</td></tr>
</table>

Object	EditCloudJobActionParamType		
Attribute Name	Type	Optional	Description
pause	String	NO	Pause a Cloud Job
resume	String	NO	Resume/Restart a Cloud Job
cancel	String	NO	Cancel a Cloud Job

Response Parameters:

SYNCHRONOUS:

Object	CloudSystemTeam		
Attribute Name	Type	Optional	Description
<CloudJob> is detailed in GET (Object)			

ASYNCHRONOUS:

Object	job		
Attribute Name	Type	Optional	Description
<Job> is detailed in the Appendix.			

2.6.2.3.1 Sample JSON request

```
{
  "action": "pause”
}
```

2.6.2.4 DELETE

Resource	cloudjob
Request Type:	DELETE
New Endpoint Description:	Deletes a Cloud Job
Modify Endpoint Description:	Added in this Release

<table>
<tr><td></td><td>
<table>
<tr><td>Not applicable</td></tr>
<tr><td>Modified in this Release</td></tr>
<tr><td>Not applicable</td></tr>
<tr><td>Removed in this Release</td></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>…/92/system/symmetrix/{symmetrixId}/cloud/cloudjob/{cloud_job_id}</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>No</td><td>Unique identifier of the symmetrix</td></tr>
<tr><td>cloud_job_id</td><td>String</td><td>No</td><td>The ID of the Cloud Job</td></tr>
</table>
</td></tr>
<tr><td>Query Parameters:</td><td>Not applicable</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>No content</td></tr>
</table>

2.6.3 storagegroup

2.6.3.1 GET (List)

<table>
<tr><td>Resource</td><td>storageggroup</td></tr>
<tr><td>Request Type:</td><td>GET</td></tr>
<tr><td>New Endpoint Description:</td><td>Gets a list of Cloud Storage Groups
A cloud Storage Group can be defined as the following:
A Local Storage Group that exists on the Array which has Cloud Snapshots
A Remote (Orphaned) Storage Group that does NOT exist on the Array which has Cloud Snapshots in the Cloud</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><td>Added in this Release</td></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
</table>

	Modified in this Release Not applicable Removed in this Release Not applicable
URL:	…/92/system/symmetrix/{symmetrixId}/cloud/storagegroup
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable
Path Parameters:	

Attribute Name	Type	Optional	Description
symmetrixId	String	No	Unique identifier of the symmetrix

Query Parameters:

Attribute Name	Type	Description
storage_group_id	String	Optional value that filters returned list to display Cloud Storage Groups infos that contain the specified Current Storage Group name or Last creation Storsage Group Name only, for example, equal to "storage_group_id=SG_NAME_123" or contain "storage_group_id=<like>123"
capacity_gb	String	Optional value that filters returned list to display Cloud Storage Group infos that contain greater than ("capacity_gb=>1"), Less than ("capacity_gb=<1") or equal to ("capacity_gb=1") the specified current capacity or last creation Capacity GB
num_of_volumes	String	Optional value that filters returned list to display Cloud Storage Group infos that contain greater than ("num_of_volumes=>1"), Less than ("num_of_volumes =<1") or equal to ("num_of_volumes =1") the specified current number of volumes or last creation number of volumes
Last_creation_date_timestamp	String	Optional value that filters returned list to display Cloud Storage Group infos that contain greater than ("last_creation_date_timestamp=>1"), Less than ("last_creation_date_timestamp=<1") or equal to ("last_creation_date_timestamp=1") the specified Last creation date timestamp

	num_of_cloud_snapshots	String	Optional value that filters returned list to display Cloud Storage Group infos that contain greater than ("num_of_cloud_snapshots=>1"), Less than ("num_of_cloud_snapshots=<1") or equal to ("num_of_cloud_snapshots=1") the specified number of cloud snapshots
	num_of_cloud_providers	String	Optional value that filters returned list to display Cloud Storge Group Infos that contain greater than ("num_of_cloud_providers=>1"), Less than ("num_of_cloud_providers=<1") or equal to ("num_of_cloud_providers=1") the specified number of cloud providers
	orphaned	String	Optional value that filters returned list to display Cloud Storage Group infos that are Orphaned (true/false)

Request Parameters:

Not applicable

Response Parameters:

Object	CloudStorageGroupListType		
Attribute Name	Type	Optional	Description
cloud_storage_group_infos	Array<CloudStorageGroupInfoType>	No	A list of cloud Storage Group infos

Object	CloudStorageGroupInfoType		
Attribute Name	Type	Optional	Description
storage_group_uuid	String	No	Unique identifier of the Cloud Storage Group
storage_group_id	String	Yes	The name of the Current Storage Group or the last creation Storage Group name
Last_creation_date_timestamp	long	Yes	The timestamp the last Snapshot was taken on the Array

2.6.3.1.1 Sample JSON response


```
{
  "cloud_storage_group_infos": [
   {
      "storage_group_uuid": "124587888",
      "storage_group_id": "SG_NAME ",
      "last_creation_date_timestamp": 1544004886
   },
  {
      "storage_group_uuid": "124587899",
      "storage_group_id": "SG_NAME_2",
      "last_creation_date_timestamp": 1544004886
   }
}
```


2.6.3.2 GET (Object)

Resource	storagegroup
Request Type:	GET
New Endpoint Description:	Gets the Cloud Storage Group Details
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL:	…/92/system/symmetrix/{symmetrixId}/cloud/storagegroup/{storage_group_uuid}
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable

Path Parameters:

Attribute Name	Type	Optional	Description
symmetrixId	String	No	Unique identifier of the symmetrix
storage_group_uuid	String	No	The Storage Group UUID

Query Parameters:	Not applicable

Request Parameters:	Not applicable
Response Parameters:	

Object	cloudStorageGroup		
Attribute Name	Type	Optional	Description
storage_group_uuid	String	No	Unique identifier of the Cloud Storage Group instance
storage_group_id	String	Yes	The name of the current storage group if it exists otherwise the last creation storage group name
capacity_gb	double	Yes	The current Capacity of the Storage Group if it exists otherwise the last creation capacity (GB)
num_of_volumes	long	Yes	The total number of volumes currently in the Storage Group if it exists otherwise the last creation number of volumes
last_creation_date	String	Yes	The date the last Snapshot was taken on the Array
last_creation_date_timestamp	long	Yes	The timestamp the last Snapshot was taken on the Array
num_of_cloud_snapshots	long	Yes	The number of Cloud Snapshots

				associated with the Storage Group
	num_of_cloud_providers	long	Yes	The number of Cloud Providers associated with the Storage Group
	orphaned	boolean	yes	Specifies if the Storage Group at the snapshot creation time still exists or not

2.6.3.2.1 Sample JSON response

```
{
        "storage_group_uuid":"4589658965478996",
        "storage_group_id":"SG_NAME_2",
        "capacity_gb":100.05,
        "num_of_volumes": 10,
        "last_creation_date":"Wed Dec 12 2018 13:13:27",
        "last_creation_date_timestamp":"4589658965475558",
        "orphaned": false,
        "num_of_cloud_snapshots":10,
        "num_of_cloud_providers":1
}
```

2.6.4 cloudsnapshot

2.6.4.1 GET (List)

Resource	cloudsnapshot
Request Type:	GET
New Endpoint Description :	Gets a List of Cloud Snapshots based on the specified Cloud Storage Group
Modify Endpoint Description :	Added in this Release

	Not applicable
	Modified in this Release
	Not applicable
	Removed in this Release
	Not applicable
URL:	…/92/system/symmetrix/{symmetrixId}/cloud/storagegroup/{storage_group_uuid}/cloudsnapshot/
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable

Path Parameters:

Attribute Name	Type	Optional	Description
symmetrixId	String	No	Unique identifier of the symmetrix
storage_group_uuid	String	No	The Storage Group unique identifier, UUID as opposed to name or id

Query Parameters:

Attribute Name	Type	Description
snapshot_name	String	Optional value that filters returned list to display Cloud Snapshots ids that contain the specified Snapshot name only, for example, equal to " snapshot_name=Weekly_123" or contain "snapshot_name=<like>123"
cloud_provider_id	String	Optional value that filters returned list to display Cloud Snapshot Ids that contain the specified Cloud Provider name only, for example, equal to "cloud_provider_id=Amazon" or contain "cloud_provider_id=<like>ama"
state	String	Optional value that filters returned list to display Cloud Snapshot Ids that contain the specified Cloud Snapshot state only, for example equal to " state=archived"
capacity_gb	String	Optional value that filters returned list to display Cloud Snapshot Ids that contain greater than ("capacity_gb=>1"), Less than

			("capacity_gb=<1") or equal to ("capacity_gb=1") the specified capacity gb
	archived_percent	String	Optional value that filters returned list to display Cloud Snapshot Ids that contain greater than ("archived_percent=>1"), Less than ("archived_percent=<1") or equal to ("archived_percent=1") the specified archived percent
	storage_group_id	String	Optional value that filters returned list to display Cloud Snapshot Ids that contain the specified Storage Group name only, for example, equal to "storage_group_id=SG_NAME" or contain "storage_group_id=<like>SG_"
	storage_group_uuid	String	Optional value that filters returned list to display Cloud Snapshot Ids that contain the specified Storage Group uuid only, for example, equal to "storage_group_uuid=154569878996542" or contain "storage_group_uuid=<like>555"
	creation_date_timestamp	String	Optional value that filters returned list to display Cloud Snapshot Ids that contain greater than ("creation_date_timestamp=>1"), Less than ("creation_date_timestamp=<1") or equal to ("creation_date_timestamp=1") the specified creation date timestamp
	expiry_date_timestamp	String	Optional value that filters returned list to display Cloud Snapshot Ids that contain greater than ("expiry_date_timestamp=>1"), Less than ("expiry_date_timestamp=<1") or equal to ("expiry_date_timestamp=1") the specified expiry date timestamp
	num_of_volumes	String	Optional value that filters returned list to display Cloud Snapshot Ids that contain greater than ("num_of_volumes=>1"), Less than ("num_of_volumes=<1") or equal to ("num_of_volumes=1") the specified number of volumes
	encryption_enabled	String	Optional value that filters returned list to display Cloud Snapshot Ids that are encrypted (true/false)
	compression_enabled	String	Optional value that filters returned list to display Cloud Snapshot Ids that are compressed (true/false)

	orphaned	String	Optional value that filters returned list to display Cloud Snapshot Ids that are Orphaned (true/false)

Request Parameters :

Not applicable

Response Parameters :

Object	CloudSnapshotListType		
Attribute Name	Type	Optional	Description
cloud_snapshot_infos	Array<CloudSnapshotInfoType>	No	A list of cloud snapshot infos

Object	CloudSnapshotInfoType		
Attribute Name	Type	Optional	Description
cloud_snapshot_id	String	No	Unique identifier of the Cloud Snaphot instance
snapshot_name	String	Yes	The name of the Cloud Snapshot which can be applied to multiple snapshots
creation_date	String	Yes	The date the Snapshot was taken on the Array
creation_date_timestamp	long	Yes	The timestamp the Snapshot was taken on the Array

2.6.4.1.1 Sample JSON response

```
{
  "cloud_snapshot_infos": [
   {
```

```
    "cloud_snapshot_id": "124587888",
    "snapshot_name": "Weekly_123",
    "creation_date": "Wed, 05 Dec 2018 10:14:46",
    "creation_date_timestamp": 1544004886
  },
  {
    "cloud_snapshot_id": "124587889",
    "snapshot_name": "Weekly_123",
    "creation_date": "Wed, 12 Dec 2018 10:14:46",
    "creation_date_timestamp": 1544609686
  }
}
```


2.6.4.2 GET (Object)

Resource	cloudsnapshot
Request Type:	GET
New Endpoint Description:	Gets the Cloud Snapshot details
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL:	…/92/system/symmetrix/{symmetrixId}/cloud/storagegroup/{storage_group_uuid}/cloudsnapshot/{cloud_snapshot_id}
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable

Path Parameters:

Attribute Name	Type	Optional	Description
symmetrixId	String	No	Unique identifier of the symmetrix
storage_group_uuid	String	No	The Storage Group unique identifier, UUID as opposed to name or id

	cloud_snapshot_id	String	No	The Cloud Snapshot Id (Not name)	
Query Parameters:	Not applicable				
Request Parameters:	Not applicable				
Response Parameters:					

Object	cloudSnapshot		
Attribute Name	Type	Optional	Description
cloud_snapshot_id	String	No	Unique identifier of the Cloud Snaphot instance
snapshot_name	String	Yes	The name of the Cloud Snapshot which can be applied to multiple snapshots
creation_date	String	Yes	The date the Snapshot was taken on the Array
creation_date_timestamp	long	Yes	The timestamp the Snapshot was taken on the Array
expiry_date	String	Yes	The date the Cloud Snapshot will expiry in the Cloud
expiry_date_timestamp	long	Yes	The Timestamp the Cloud Snapshot will expiry in the Cloud
state	String	Yes	The Current State of the Cloud Snapshot
cloud_provider_id	String	Yes	The Name of the associated Cloud Provider

capacity_gb	double	Yes	The Total capacity of the Storage Group at the snapahot creation time	
archived_percent	long	Yes	The current progress percent of the snapshot being backed up to the cloud where applicable	
storage_group_id	String	Yes	The Name of the Storage Group at the time the Snapshot was created	
storage_group_uuid	String	Yes	The unique identifier of the Storage Group	
num_of_volumes	long	Yes	The number of Volume in the Storage Group at the time the snapshot was created	
encryption_enabled	boolean	Yes	Specifies if encryption is enabled or not	
compression_enabled	boolean	yes	Specifies if Compression is enabled or not	
orphaned	boolean	yes	Specifies if the Storage Group at the snapshot creation time still exists or not	

2.6.4.2.1 Sample JSON response


```
{
  "cloud_snapshot_id": "124587889",
  "snapshot_name": "Weekly_123",
  "creation_date": "Wed, 12 Dec 2018 10:14:46",
  "creation_date_timestamp": 1544609686,
  "expiry_date": "Wed, 12 Jun 2019 10:14:46",
  "expiry_date_timestamp": 1560334486,
  "state ": "Archivied",
  "cloud_provider_name ": "Archivied",
  "capacity_gb ": 10.25,
```

```
  "archived_percent ": 100
  "storage_group_id ": "SG_NAME",
  "storage_group_uuid ": 4587954785965874,
  “num_of_volumes”:3,
  "encryption_enabled": true,
  "compression_enabled ": true,
  "orphaned ": false
}
```

2.6.4.3 PUT

Resource	cloudsnapshot
Request Type:	PUT
New Endpoint Description:	Modifies the Cloud Snapshot
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL:	…/92/system/symmetrix/{symmetrixId}/cloud/storagegroup/{storage_group_uuid}/cloudsnapshot/{cloud_snapshot_id}
User Roles:	Administrator/Storage Administrator
Prerequisite:	Not applicable
Path Parameters:	Attribute Name | Type | Optional | Description symmetrixId | String | No | Unique identifier of the symmetrix

	storage_group_uuid	String	No	The Storage Group unique identifier, UUID as opposed to name or id	
	cloud_snapshot_id	String	No	The Cloud Snapshot Id (Not name)	

Query Parameters:

Not applicable

Request Parameters:

Object	EditCloudSnapshotParam		
Attribute Name	Type	Optional	Description
edit_cloud_snapshot_action_param	EditCloudSnapshotActionParamType	No	Select one of the Edit Options to modify the specified Cloud Snapshot

Object	EditCloudSnapshotActionParamType		
Attribute Name	Type	Optional	Description
action	Enum (CloudSnapshotAction) [pause, resume, cancel, recover, alter]	No	Specifies the action
pause	PauseCloudSnapshotParamType	Yes	Pauses the specified Cloud Snapshot in the following states only “Archiving”
resume	ResumeCloudSnapshotParamType	Yes	Pauses the specified Cloud Snapshot in the following states only “Paused”

cancel	CancelCloudSnapshotParamType	Yes	Cancels the specified Cloud Snapshot in the following states only “Created, Queued, Archiving”
recover	RecoverCloudSnapshotParamType	Yes	Creates a local copy of the specified Cloud Snapshot in the following states only “Archived”

Object	PauseCloudSnapshotParamType		
Attribute Name	Type	Optional	Description
Not Applicable	Not Applicable	Not Applicable	Not Applicable

Object	ResumeCloudSnapshotParamType		
Attribute Name	Type	Optional	Description
Not Applicable	Not Applicable	Not Applicable	Not Applicable

Object	CancelCloudSnapshotParamType		
Attribute Name	Type	Optional	Description
Not Applicable	Not Applicable	Not Applicable	Not Applicable

Object	RecoverCloudSnapshotParamType		
Attribute Name	Type	Optional	Description
storage_group_id	String	No	The New Storage Group Name.

Object	AlterCloudSnapshotParamType

	<table><tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr><tr><td>new_expiration_date</td><td>String</td><td>No</td><td>The New Expiration Date.</td></tr></table>
Response Parameters:	SYNCHRONOUS: <table><tr><th>Object</th><td colspan="3">cloudSnapshot</td></tr><tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr><tr><td colspan="4"><Cloud Snapshot> is detailed in GET (Object)</td></tr></table> ASYNCHRONOUS: <table><tr><th>Object</th><td colspan="3">job</td></tr><tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr><tr><td colspan="4"><Job> is detailed in the Appendix.</td></tr></table>

2.6.4.3.1 Sample JSON request

```
{
  "edit_cloud_snapshot_action_param": {
    "recover": {
     “storage_group_id”: ”NEW_SG_NAME”
     }
  }
}
```

2.6.4.4 DELETE

Resource	cloudsnapshot
Request Type:	DELETE
New Endpoint Description:	Deletes a Cloud Snapshot when in the following states only: • Created • Queued • Archived
Modify Endpoint Description:	<table><tr><th>Added in this Release</th></tr><tr><td>Not applicable</td></tr><tr><th>Modified in this Release</th></tr><tr><td>Not applicable</td></tr><tr><th>Removed in this Release</th></tr></table>

<table>
<tr><td></td><td>Not applicable</td></tr>
<tr><td>URL:</td><td>…/92/system/symmetrix/{symmetrixId}/cloud/storagegroup/{storage_group_uuid}/cloudsnapshot/{cloud_snapshot_id}</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>No</td><td>Unique identifier of the symmetrix</td></tr>
<tr><td>storage_group_uuid</td><td>String</td><td>No</td><td>The Storage Group unique identifier, UUID as opposed to name or id</td></tr>
<tr><td>cloud_snapshot_id</td><td>String</td><td>No</td><td>The Cloud Snapshot Id (Not name)</td></tr>
</table>
</td></tr>
<tr><td>Query Parameters:</td><td>Not applicable</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>No content</td></tr>
</table>

2.6.5 cloudprovider

2.6.5.1 GET (List)

<table>
<tr><td>Resource</td><td>cloudprovider</td></tr>
<tr><td>Request Type:</td><td>GET</td></tr>
<tr><td>New Endpoint Description:</td><td>Gets a list of Cloud Provider names</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
</table>

	Modified in this Release
	Not applicable
	Removed in this Release
	Not applicable
URL:	… /92/system/symmetrix/{symmetrixId}/cloud/cloudprovider
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable

Path Parameters:

Attribute Name	Type	Description
symmetrixId	String	Unique identifier of the symmetrix

Query Parameters:

Attribute Name	Type	Description
cloudprovider_id	String	Optional value that filters returned list to display Cloud Provider names that contain the specified Cloud Provider name only, for example, equal to "cloudprovider_id=Amazon" or contain "cloudprovider_id =<like>Am"
provider	String	Optional value that filters returned list to display Cloud Provider names that contain the specified Cloud Provider name only, for example, equal to "cloudprovider_id=Amazon" or contain "cloudprovider_id =<like>Ama"
state	String	Optional value that filters returned list to display Cloud Provider names that contain the specified Cloud Provider state only, for example equal to " state=online"
uuid	String	Optional value that filters returned list to display Cloud Provider names that contain the specified UUID name only, for example, equal to "uuid=Amazon" or contain "uuid=<like>Ama"
num_of_storage_groups	String	Optional value that filters returned list to display Cloud Provider names that contain greater than("number_of_storage_groups =>1"),

			Less than ("number_of_storage_groups =<1") or equal to ("number_of_storage_groups =1") the specified number of Storage Groups
	num_of_cloud_snapshots	String	Optional value that filters returned list to display Cloud Provider names that contain greater than("number_of_cloud_snapshots=>1"), Less than ("number_of_cloud_snapshots=<1") or equal to ("number_of_cloud_snapshots=1") the specified number of cloud snapshots

Request Parameters:	Not applicable			
Response Parameters:	Object	ListCloudProviderResult		
	Attribute Name	Type	Optional	Description
	Cloudprovider_ids	String array	No	A list of cloud provider names

2.6.5.1.1 Sample JSON response

```
{
  "cloudprovider_ids": [
  "Amazon Cloud",
 “Azure”,
 “G1”
}
```

2.6.5.2 GET (Object)

Resource	cloudprovider
Request Type:	GET
New Endpoint Description :	Gets a Cloud Provider details
Modify Endpoint Description :	Added in this Release: Not applicable

	Modified in this Release
	Not applicable
	Removed in this Release
	Not applicable
URL:	… /92/system/symmetrix/{symmetrixId}/cloud/cloudprovider/{cloud_provider_id}
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable

Path Parameters:

Attribute Name	Type	Optional	Description
symmetrixId	String	No	Unique identifier of the symmetrix
cloud_provider_id	String	No	The Cloud Provider Name

Query Parameters: Not applicable

Request Parameters: Not applicable

Response Parameters:

Object	cloudProvider		
Attribute Name	Type	Optional	Description
cloud_provider_id	String	No	The name of the Cloud Provider
type	String	Yes	The Type of Cloud Provider, example ECS
state	String	Yes	The current state of the Cloud Provider
uuid	String	Yes	The unique identifier

			of the Cloud Provider
used_capacity_gb	Double	Yes	The Total Cloud Provider Used Capacity in GB
num_of_cloud_storage_groups	int	Yes	The number of Cloud Storage Groups associated with the Cloud Proivder
num_of_cloud_snapshots	int	Yes	The number of Cloud Snapshots associated with the Cloud Proivder
num_of_cloud_policies	int	Yes	The number of Cloud Policies associated with the Cloud Proivder
cloud_provider_connection_details	CloudProviderConnectionDetailsType	yes	The Cloud Provider connection details associated with each of the types

Object	CloudProviderConnectionDetailsType		
Attribute Name	Type	Optional	Description

azure_connection_details	AzureConnectionDetailsType	Yes	Azure specific Details
amazon_connection_details	AmazonConnectionDetailsType	Yes	Amazon specific Details
ecs_connection_details	EcsConnectionDetailsType	Yes	Ecs specific Details

Object	AmazonConnectionDetailsType		
Attribute Name	Type	Optional	Description
key	String	No	AWSAccessKeyId
secret	String	Yes	AWSSecretAccessKey
secure	Boolean	Yes	Use HTTPS. Valid values are "true" and "false"
node	String	Yes	Amazon Node Name
port	String	Yes	Port
storage_class	String	Yes	Storage Class
region	String	Yes	Amazon Region

Object	AzureConnectionDetailsType		
Attribute Name	Type	Optional	Description
storage_account	String	No	Storage Account
managed_key	String	Yes	Managed Key
secure	Boolean	Yes	Use HTTPS. Valid values are "true" and "false"
url	String	Yes	Url
port	String	Yes	Port
container	String	Yes	container

Object	ECSConnectionDetailsType		
Attribute Name	Type	Optional	Description

	key	String	No	Key
	secret	String	Yes	Secret
	secure	Boolean	Yes	Use HTTPS. Valid values are "true" and "false"
	url	String	Yes	Url
	port	String	Yes	Port
	bucket	String	Yes	bucket
	resquest_style	String	Yes	Request style to be used. Valid values are "auto", "virtual" and "path". Defaults to auto

2.6.5.2.1 Sample JSON response

```
{
  "cloud_provider_id": "Azure",
  "uuid": "05a014ca-da15-11e9-86a2-49850db0977b",
  "type": "azure",
  "state": "online",
  “used_capacity_gb”:120.25,
  “num_of_cloud_storage_groups”:3,
  “num_of_cloud_snapshots”:30,
  “num_of_cloud_policies”:3,
  "azure_connection_details": {
    "storage_account": "AZURE_AC",
    "url": "azure.test.url",
    "secure": true,
    "port": "443",
    "container": “CONTAINER_NAME”
  }
}
```

2.6.5.3 POST

Resource	cloud
Request Type:	POST
New Endpoint Description :	Create a Cloud Provider
Modify Endpoint Description :	Added in this Release Not applicable

	Modified in this Release Not applicable Removed in this Release Not applicable
URL:	… /92/system/symmetrix/{symmetrixId}/cloud/cloudprovider
User Roles:	Administrator/Storage Administrator
Prerequisite:	Not applicable

Path Parameters:

Attribute Name	Type	Description
symmetrixId	String	Unique identifier of the symmetrix

Query Parameters: Not applicable

Request Parameters:

Object	createCloudProviderParam		
Attribute Name	Type	Optional	Description
cloud_provider_id	String	No	Unique identifier of the Cloud Provider instance
cloud_provider_connection_details	CloudProviderConnectionDetails Type (see GET (Object) for each cloud provider details)	yes	The Cloud Provider connection details associated with each of the types

Response Parameters:

SYNCHRONOUS:

Object	cloudProvider		
Attribute Name	Type	Optional	Description

	(see GET (Object) for cloud provider details) ASYNCHRONOUS: Object: job Attribute Name / Type / Optional / Description <Job> is detailed in the Appendix.

2.6.5.3.1 Sample JSON request

```
{
    “cloud_provider_id":"Azure",
    "cloud_provider_connection_details”: {
      "azure_connection_details": {
          "storage_account": "AZURE_AC",
        "url": "azure.test.url",
        "secure": true,
        "port": "443",
        "managed_key": “KEY”
      }
  }
}
```

2.6.5.4 DELETE

Resource	cloudprovider
Request Type:	DELETE
New Endpoint Description:	Delete a Cloud Provider
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL:	… /92/system/symmetrix/{symmetrixId}/cloudprovider/{cloud_provider_id}
User Roles:	Administrator/Storage Administrator
Prerequisite:	Not applicable

<table>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>No</td><td>Unique identifier of the symmetrix</td></tr>
<tr><td>cloud_provider_id</td><td>String</td><td>No</td><td>The Cloud Provider Name</td></tr>
</table>
</td></tr>
<tr><td>Query Parameters:</td><td>Not applicable</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>No content</td></tr>
</table>

2.6.6 backup

2.6.6.1 POST

<table>
<tr><td>Resource</td><td>backup</td></tr>
<tr><td>Request Type:</td><td>POST</td></tr>
<tr><td>New Endpoint Description:</td><td>Creates a downloadable backup file of the current Cloud System on the specified Symmmetrix</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>…/92/system/symmetrix/{symmetrixId}/cloud/backup</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>No</td><td>Unique identifier of the symmetrix</td></tr>
</table>
</td></tr>
<tr><td>Query Parameters:</td><td>Not applicable</td></tr>
</table>

<table>
<tr><td>Request Parameters:</td><td>
<table>
<tr><td>Object</td><td colspan="3">CloudSystemBackupParam</td></tr>
<tr><td>Attribute Name</td><td>Type</td><td>Optional</td><td>Description</td></tr>
<tr><td>password</td><td>String</td><td>No</td><td>The Password for the gzip file</td></tr>
</table>
</td></tr>
<tr><td>Response Parameters:</td><td>
<table>
<tr><td>Object</td><td>gzip file</td></tr>
</table>
</td></tr>
</table>

2.6.7 networking

2.6.7.1 GET (Object)

<table>
<tr><td>Resource</td><td>networking</td></tr>
<tr><td>Request Type:</td><td>GET</td></tr>
<tr><td>New Endpoint Description:</td><td>Gets the Network summary details</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><td>Added in this Release</td></tr>
<tr><td>Not applicable</td></tr>
<tr><td>Modified in this Release</td></tr>
<tr><td>Not applicable</td></tr>
<tr><td>Removed in this Release</td></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>…/92/system/symmetrix/{symmetrixId}/cloud/networking/</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
</table>

Path Parameters:

Attribute Name	Type	Optional	Description
symmetrixId	String	No	Unique identifier of the symmetrix

Query Parameters: Not applicable

Request Parameters: Not applicable

Response Parameters:

Object	CloudSystemNetworkingResult		
Attribute Name	Type	Optional	Description
num_of_interfaces	Integer	yes	The total number of interfaces
num_of_teams	Integer	yes	The total number of interfaces
num_of_routes	Integer	yes	The total number of interfaces
num_of_dns_servers	Integer	yes	The total number of interfaces
bandwidth_throttling	BandwidthThrottling	Yes	Gets the value of the bandwidth_Throttling

Object	BandwidthThrottling		
Attribute Name	Type	Optional	Description
enable	Boolean	Not	Defines if Bnadwidth limits are set
bandwidth_limit_mbps	Long	yes	The Bandwidth Limit in mbps

2.6.7.1.1 Sample JSON response

```
{
   "num_of_interfaces": 10,
  “num_of_teams”:1,
  “num_of_routes”:3,
  “num_of_dns_servers”:1
   “bandwidth_throttling”: {
        “enable”: true,
        “bandwidth_limit_mbps”: 100
    }
}
```

2.6.7.2 PUT

Resource	networking
Request Type:	PUT
New Endpoint Description:	This call modifies The Cloud System General Networking settings
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL:	…/92/system/symmetrix/{symmetrixId}/cloud/networking/
User Roles:	Administrator/Storage Administrator
Prerequisite:	Not applicable
Path Parameters:	Attribute Name: symmetrixId; Type: String; Optional: No; Description: Unique identifier of the symmetrix
Query Parameters:	Not applicable
Request Parameters:	Object: EditCloudSystemNetworkParam Attribute Name: edit_cloud_system_networking_action; Type: EditCloudSystemNetworkingActionParam; Optional: No; Description: Select one of the Edit

			Options to modify the specified Cloud System Network

Object	EditCloudSystemNetworkingActionParamType		
Attribute Name	Type	Optional	Description
edit_bandwidth_limit	EditBandwidthLimitParamType	YES	Sets/removes a Bandwidth limit on the Cloud system. .

Object	EditBandwidthLimitParamType		
Attribute Name	Type	Optional	Description
enable	Boolean	NO	Enable/Disable the bandwidth limits
bandwidth_limit_mbps	Interger	NO	defines the limit in mbs per second

Response Parameters:

SYNCHRONOUS:

Object	CloudSystem		
Attribute Name	Type	Optional	Description
<CloudSystem> is detailed in GET (Object)			

ASYNCHRONOUS:

Object	job

	Attribute Name	Type	Optional	Description
	<Job> is detailed in the Appendix.			

2.6.7.2.1 Sample JSON request

```
{
  "edit_cloud_system_networking_action": {
    "edit_bandwidth_limit": {
        “enable”:true,
        “bandwidth_limit_mbps”:300
    }
  }
}
```

2.6.8 dns

2.6.8.1 GET (List)

Resource	dns
Request Type:	GET
New Endpoint Description:	Gets a List of Cloud Sytem DNS servers
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL:	…/92/system/symmetrix/{symmetrixId}/cloud/networking/dns
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable
Path Parameters:	Attribute Name / Type / Optional / Description symmetrixId / String / No / Unique identifier of the symmetrix

<table>
<tr><td>Query Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr>
<tr><td>dns_id</td><td>String</td><td>Optional value that filters returned list to display Cloud System DNS servers that have a specified dns_id example equal to "dns_id=10.60.122.121" or contains "dns_id=<like>10.20"</td></tr>
</table>
</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>
<table>
<tr><th>Object</th><td colspan="3">ListCloudSystemDnsServersResultType</td></tr>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>dns_id</td><td>Array<String></td><td>No</td><td>A list of dns servers</td></tr>
</table>
</td></tr>
</table>

2.6.8.1.1 Sample JSON response

```
{
  "dns_id": [
     "10.60.122.25",
     "10.60.122.26"
   ]
}
```

2.6.8.2 GET (Object)

<table>
<tr><td>Resource</td><td>dns</td></tr>
<tr><td>Request Type:</td><td>GET</td></tr>
<tr><td>New Endpoint Description:</td><td>Gets the DNS server Details</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
</table>

	Removed in this Release Not applicable
URL:	…/92/system/symmetrix/{symmetrixId}/cloud/networking/dns/{dns_id}
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable

Path Parameters:

Attribute Name	Type	Optional	Description
symmetrixId	String	No	Unique identifier of the symmetrix
dns_id	String	No	The DNS server Id

Query Parameters:	Not applicable
Request Parameters:	Not applicable

Response Parameters:

Object	CloudSystemDnsServer		
Attribute Name	Type	Optional	Description
dns_id	String	No	The DNS server id (IP Address)

2.6.8.2.1 Sample JSON response

```
{
  "dns_id": "10.60.120.125"
}
```

2.6.8.3 POST

Resource	dns
Request Type:	POST
New Endpoint Description:	Adds a DNS Server to the Cloud System
Modify Endpoint Description:	Added in this Release Not applicable

	Modified in this Release Not applicable Removed in this Release Not applicable
URL:	… /92/system/symmetrix/{symmetrixId}/networking/dns
User Roles:	Administrator/Storage Administrator
Prerequisite :	Not applicable

Path Parameters:

Attribute Name	Type	Description
symmetrixId	String	Unique identifier of the symmetrix

Query Parameters: Not applicable

Request Parameters:

Object	CreateCloudSystemDnsServerParam		
Attribute Name	Type	Optional	Description
ip_address	String	No	The IP Address of the DNS Server

Response Parameters:

SYNCHRONOUS:

Object	CloudSystemDnsServer		
Attribute Name	Type	Optional	Description
(see GET (Object) for cloud details)			

ASYNCHRONOUS:

Object	job		
Attribute Name	Type	Optional	Description
<Job> is detailed in the Appendix.			

2.6.8.3.1 Sample JSON request

```
{
 “ip_address” : “1.1.1.1”
}
```

2.6.8.4 PUT

<table>
<tr><td>Resource</td><td>dns</td></tr>
<tr><td>Request Type:</td><td>PUT</td></tr>
<tr><td>New Endpoint Description:</td><td>Modifies the DNS server</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>…/92/system/symmetrix/{symmetrixId}/cloud/networking/dns/{dns_id}</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>No</td><td>Unique identifier of the symmetrix</td></tr>
<tr><td>dns_id</td><td>String</td><td>No</td><td>The Cloud System DNS Server Id</td></tr>
</table>
</td></tr>
<tr><td>Query Parameters:</td><td>Not applicable</td></tr>
<tr><td>Request Parameters:</td><td>
<table>
<tr><th>Object</th><td>EditCloudSystemDnsServerParam</td></tr>
</table>
</td></tr>
</table>

Attribute Name	Type	Optional	Description
edit_cloud_system_dns_server_action	EditCloudSystemDnsServerActionParamType	No	Select one of the Edit Options to modify the specified Cloud System DNS Server

Object	EditCloudSystemDnsServerActionParamType		
Attribute Name	Type	Optional	Description
edit_cloud_system_dns_server_ip_address	EditCloudSystemDnsServerIpAddressParamType	YES	Edits the DNS Server IP Address

Object	EditCloudSystemDnsServerIpAddressParamType		
Attribute Name	Type	Optional	Description
ip_address	String	NO	The IP Address of the DNS Server

Response Parameters:

SYNCHRONOUS:

Object	CloudSystemDnsServer		
Attribute Name	Type	Optional	Description
<CloudSystemDnsServer> is detailed in GET (Object)			

ASYNCHRONOUS:

	Object	job		
	Attribute Name	Type	Optional	Description
	<Job> is detailed in the Appendix.			

2.6.8.4.1 Sample JSON request

```
{
  "edit_cloud_system_dns_server_action": {
    "edit_cloud_system_dns_server_ip_address": {
     “ip_address”:”10.10.10.10”
     }
  }
}
```

2.6.8.5 DELETE

Resource	dns
Request Type:	DELETE
New Endpoint Description:	Deletes a Cloud System DNS Server
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL:	…/92/system/symmetrix/{symmetrixId}/cloud/networking/dns/{dns_id}
User Roles:	Administrator/Storage Administrator
Prerequisite:	Not applicable
Path Parameters:	Attribute Name / Type / Optional / Description symmetrixId / String / No / Unique identifier of the symmetrix dns_id / String / No / The Cloud System DNS Server Id
Query Parameters:	Not applicable
Request Parameters:	Not applicable

Response Parameters:	No content

2.6.9 interface

2.6.9.1 GET (List)

Resource	interface
Request Type:	GET
New Endpoint Description:	Gets a List of Cloud Sytem Network Interfaces
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL:	…/92/system/symmetrix/{symmetrixId}/cloud/networking/interface
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable

Path Parameters:

Attribute Name	Type	Optional	Description
symmetrixId	String	No	Unique identifier of the symmetrix

Query Parameters:

Attribute Name	Type	Description
ip_address	String	Optional value that filters returned list to display Cloud System Interfaces that have a specified ip_address example equal to "ip_address=10.60.122.121" or contains " ip_address=<like>10.20"

	state	String	Optional value that filters returned list to display Cloud System Interfaces that have a specified state example equal to "state=online" or contains "state=<like>off"
	prefix	Integer	Optional value that filters returned list to display Cloud System Interfaces that contain greater than ("prefix=>1"), Less than ("prefix=<1") or equal to ("prefix=1")
	mac_addresss	String	Optional value that filters returned list to display Cloud System Interfaces that have a specified mac_addresss example: equal to " mac_addresss=10.60.122.121" or contains "mac_addresss =<like>10.20"
	mtu	Integer	Optional value that filters returned list to display Cloud System Interfaces that contain greater than ("mtu=>1"), Less than ("mtu=<1") or equal to ("mtu=1")
	num_of_routes	Integer	Optional value that filters returned list to display Cloud System Interfaces that contain greater than ("num_of_routes =>1"), Less than ("num_of_routes =<1") or equal to ("num_of_routes =1")
	type	String	Optional value that filters returned list to display Cloud System Interfaces that have a specified dns_id example: equal to "dns_id=type" or contains "dns_id=<like>ty"
	ip_source	String	Optional value that filters returned list to display Cloud System Interfaces that have a specified ip_source example: equal to "ip_source=10.60.122.121" or contains "ip_source=<like>10.20"

Request Parameters: Not applicable

Response Parameters:

Object	ListCloudSystemInterfaceResult		
Attribute Name	Type	Optional	Description
interface_id	Array<String>	No	A list of Interfaces idss

2.6.9.1.1 Sample JSON response

```
{
  "interface_id": [
     "eth0",
     "eth1"
   ]
}
```

2.6.9.2 GET (Object)

Resource	interface
Request Type:	GET
New Endpoint Description:	Gets the network interface Details
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL:	…/92/system/symmetrix/{symmetrixId}/cloud/networking/interface/{interface_id}
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable

Path Parameters:

Attribute Name	Type	Optional	Description
symmetrixId	String	No	Unique identifier of the symmetrix
interface_id	String	No	The Interface Id

Query Parameters:	Not applicable
Request Parameters:	Not applicable

Response Parameters:

Object	CloudSystemInterface		
Attribute Name	Type	Optional	Description

	Interface_id	String	No	The network interface name
	State	String	Yes	The state of the interface
	ip_address	String	Yes	The IP address set on the Interface
	prefix	Integer	Yes	The netmask prefix set on the Interface
	mac_addresss	String	Yes	The Network interface MAC address
	mtu	Integer	Yes	The Network interface MTU
	num_of_routes	Integer	Yes	The number of Routes associated with the Network interface
	type	String	Yes	The Network interface type
	ip_source	String	Yes	The Network interface IP source

2.6.9.2.1 Sample JSON response


```
{
   "interface_id": "eth0",
   "state": "up",
   "ip_address": "10.60.141.165",
   "prefix": 23,
   "mac_addresss": "00:50:56:a6:d9:81",
   "mtu": 1500,
   "num_of_routes": 0,
   "type": "ethernet",
   "ip_source": "static"
}
```


2.6.9.3 PUT

Resource	interface

<table>
<tr><td>Request Type:</td><td>PUT</td></tr>
<tr><td>New Endpoint Description:</td><td>Modifies the Network Interface</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>…/92/system/symmetrix/{symmetrixId}/cloud/networking/interface/{interface_id}</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>No</td><td>Unique identifier of the symmetrix</td></tr>
<tr><td>interface_id</td><td>String</td><td>No</td><td>The Cloud System Network Interface Id</td></tr>
</table>
</td></tr>
<tr><td>Query Parameters:</td><td>Not applicable</td></tr>
<tr><td>Request Parameters:</td><td>
<table>
<tr><th>Object</th><th colspan="3">EditCloudSystemInterfaceParam</th></tr>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>edit_cloud_system_interface_action</td><td>EditCloudSystemInterfaceActionParamType</td><td>No</td><td>Select one of the Edit Options to modify the specified</td></tr>
</table>
</td></tr>
</table>

			Cloud System Network Interface

Object	EditCloudSystemInterfaceActionParamType		
Attribute Name	Type	Optional	Description
edit_interface_ip_settings	EditInterfaceIpSettingsParamType	YES	defines the new ip settings for the Cloud System Interface.

Object	EditInterfaceIpSettingsParamType		
Attribute Name	Type	Optional	Description
ip_address	String	NO	The IP Address of the Interface
prefix	Interger	NO	The Netmask Prefix of the Interface

Response Parameters:

SYNCHRONOUS:

Object	CloudSystemInterface		
Attribute Name	Type	Optional	Description
<CloudSystemInterface> is detailed in GET (Object)			

ASYNCHRONOUS:

Object	job		
Attribute Name	Type	Optional	Description
<Job> is detailed in the Appendix.			

2.6.9.3.1 Sample JSON request

```
{
  "edit_cloud_system_interface_action": {
    "edit_interface_ip_settings": {
        “ip_address”:”10.10.10.10”,
        “prefix”:23
    }
  }
}
```

2.6.10 route

2.6.10.1 GET (List)

Resource	route
Request Type:	GET
New Endpoint Description:	Gets a List of Cloud Sytem Routes
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL:	…/92/system/symmetrix/{symmetrixId}/cloud/networking/route
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable
Path Parameters:	Attribute Name: symmetrixId; Type: String; Optional: No; Description: Unique identifier of the symmetrix
Query Parameters:	Attribute Name: interface_id; Type: String; Description: Optional value that filters returned list to display Cloud System Routes that have a

			specified interface_id example: equal to "interface_id=eth0" or contains “interface_id=<like>0"

Request Parameters:	Not applicable

Response Parameters:

Object	ListCloudSystemRouteResultType		
Attribute Name	Type	Optional	Description
route_id	Array<String>	No	A list of Cloud System Routes

2.6.10.1.1 Sample JSON response

```
{
  "route_id": [
     "10.60.122.25-23-1.1.1.1"
  ]
}
```

2.6.10.2 GET (Object)

Resource	route
Request Type:	GET
New Endpoint Description:	Gets the Route Details
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL:	…/92/system/symmetrix/{symmetrixId}/cloud/networking/route/{route_id}
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor

Prerequisite: Not applicable

Path Parameters:

Attribute Name	Type	Optional	Description
symmetrixId	String	No	Unique identifier of the symmetrix
route_id	String	No	The Route Id

Query Parameters: Not applicable

Request Parameters: Not applicable

Response Parameters:

Object	CloudSystemRoute		
Attribute Name	Type	Optional	Description
route_id	CloudSystemRouteIdType	No	The route id: GatewayIP-prefix-DetinationIP
Interface_id	CloudSystemInterfaceIdType	Yes	The asssoicated interface Id
destination_ip_address	String	Yes	The destination IP Address of the Cloud System Route
prefix	String	Yes	The Netmask Prefix Number of the Cloud System Route
gateway_ip_address	String	Yes	The Gateway IP Address of the Cloud System Route

2.6.10.2.1 Sample JSON response


```
{
   "route_id": "default---10.23.23.2",
   "destination_ip_address": "default",
   "prefix": "-",
   "gateway_ip_address": "10.23.23.2",
   "interface_id": "eth1"
}
```


2.6.10.3 POST

Resource	route
Request Type:	POST
New Endpoint Description:	Create a Cloud System Network Route
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL:	… /92/system/symmetrix/{symmetrixId}/networking/route
User Roles:	Administrator/Storage Administrator
Prerequisite:	Not applicable

Path Parameters:

Attribute Name	Type	Description
symmetrixId	String	Unique identifier of the symmetrix

Query Parameters:	Not applicable

Request Parameters:

Object	CreateCloudSystemRouteParam		
Attribute Name	Type	Optional	Description

destination_ip_address	String	No	The Destination IP Address of the Route
gateway_ip_address	String	No	The Gateway IP Address of the Route
prefix	Integer	No	The Netmask prefix of the Route

Attribute Name	Type	Optional	Description
Interface_id	String	No	The Interface Id to be Associated with the Route
team_id	String	No	The Team Id to be Associated with the Route

Response Parameters:

SYNCHRONOUS:

Object	CloudSystemRoute		
Attribute Name	Type	Optional	Description
(see GET (Object) for cloud details)			

ASYNCHRONOUS:

Object	job		
Attribute Name	Type	Optional	Description
<Job> is detailed in the Appendix.			

2.6.10.3.1 Sample JSON request

```
{
```

```
    “destination_ip_address”:”1.1.1.1”,
    “destination_ip_address”:”2.2.2.2”,
    “prefix”: 23,
    “Interface_id”:”eth0”
}
```

2.6.10.4 DELETE

Resource	route
Request Type:	DELETE
New Endpoint Description:	Deletes a Cloud System Route
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL:	…/92/system/symmetrix/{symmetrixId}/cloud/networking/route/{route_id}
User Roles:	Administrator/Storage Administrator
Prerequisite:	Not applicable
Path Parameters:	Attribute Name / Type / Optional / Description symmetrixId / String / No / Unique identifier of the symmetrix route_id / String / No / The Cloud System Route Id
Query Parameters:	Not applicable
Request Parameters:	Not applicable
Response Parameters:	No content

2.6.11 team

2.6.11.1 GET (List)

<table>
<tr><td>Resource</td><td>team</td></tr>
<tr><td>Request Type:</td><td>GET</td></tr>
<tr><td>New Endpoint Description:</td><td>Gets a List of Cloud Sytem NIC Teams</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>…/92/system/symmetrix/{symmetrixId}/cloud/networking/team</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>No</td><td>Unique identifier of the symmetrix</td></tr>
</table>
</td></tr>
<tr><td>Query Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr>
<tr><td>ip_address</td><td>String</td><td>Optional value that filters returned list to display Cloud System Teams that have a specified ip_address example: equal to "ip_address=10.60.122.121" or contains " ip_address=<like>10.20"</td></tr>
<tr><td>state</td><td>String</td><td>Optional value that filters returned list to display Cloud System Teams that have a specified state example: equal to "state=online" or contains "state=<like>off"</td></tr>
<tr><td>prefix</td><td>Integer</td><td>Optional value that filters returned list to display Cloud System Teams that contain greater than ("prefix=>1"), Less than ("prefix=<1") or equal to ("prefix=1")</td></tr>
</table>
</td></tr>
</table>

	mac_addresss	String	Optional value that filters returned list to display Cloud System Teams that have a specified mac_addresss example: equal to " mac_addresss=10.60.122.121" or contains "mac_addresss =<like>10.20"
	mtu	Integer	Optional value that filters returned list to display Cloud System Teams that contain greater than ("mtu=>1"), Less than ("mtu=<1") or equal to ("mtu=1")
	num_of_routes	Integer	Optional value that filters returned list to display Cloud System Teams that contain greater than ("num_of_routes =>1"), Less than ("num_of_routes =<1") or equal to ("num_of_routes =1")
	type	String	Optional value that filters returned list to display Cloud System Teams that have a specified dns_id example: equal to "dns_id=type" or contains "dns_id=<like>ty"
	ip_source	String	Optional value that filters returned list to display Cloud System Teams that have a specified ip_source example: equal to "ip_source=10.60.122.121" or contains "ip_source=<like>10.20"

Request Parameters: Not applicable

Response Parameters:

Object	ListCloudSystemTeamResult		
Attribute Name	Type	Optional	Description
team_id	Array<String>	No	A list of Team ids

2.6.11.1.1 Sample JSON response

```
{
  "team_id": [
     "team0"
  ]
}
```

2.6.11.2 GET (Object)

<table>
<tr><td>Resource</td><td>team</td></tr>
<tr><td>Request Type:</td><td>GET</td></tr>
<tr><td>New Endpoint Description:</td><td>Gets the NIC Team Details</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>…/92/system/symmetrix/{symmetrixId}/cloud/networking/team/{team_id}</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>No</td><td>Unique identifier of the symmetrix</td></tr>
<tr><td>team_id</td><td>String</td><td>No</td><td>The Team Id</td></tr>
</table>
</td></tr>
<tr><td>Query Parameters:</td><td>Not applicable</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>
<table>
<tr><th>Object</th><th colspan="3">CloudSystemTeam</th></tr>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>team_id</td><td>String</td><td>No</td><td>The network Team name</td></tr>
<tr><td>State</td><td>String</td><td>Yes</td><td>The state of the Team</td></tr>
<tr><td>ip_address</td><td>String</td><td>Yes</td><td>The IP address set on the Team</td></tr>
</table>
</td></tr>
</table>

	prefix	Integer	Yes	The netmask prefix set on the Interf Team ace
	mac_addresss	String	Yes	The Network Team MAC address
	mtu	Integer	Yes	The Network Team MTU
	num_of_routes	Integer	Yes	The number of Routes associated with the Network Team
	type	String	Yes	The Network Team type
	ip_source	String	Yes	The Network Team IP source

2.6.11.2.1 Sample JSON response


```
{
    "team_id": "team0",
    "state": "up",
    "ip_address": "10.60.141.165",
    "prefix": 23,
    "mac_addresss": "00:50:56:a6:d9:81",
    "mtu": 1500,
    "num_of_routes": 0,
    "type": "ethernet",
    "ip_source": "static"
}
```


2.6.11.3 PUT

Resource	team
Request Type:	PUT
New Endpoint Description:	Modifies the Network Team
Modify Endpoint Description:	Added in this Release

	Not applicable
	Modified in this Release
	Not applicable
	Removed in this Release
	Not applicable
URL:	…/92/system/symmetrix/{symmetrixId}/cloud/networking/team/{team_id}
User Roles:	Administrator/Storage Administrator
Prerequisite:	Not applicable

Path Parameters:

Attribute Name	Type	Optional	Description
symmetrixId	String	No	Unique identifier of the symmetrix
team_id	String	No	The Cloud System Network Team Id

Query Parameters: Not applicable

Request Parameters:

Object	EditCloudSystemTeamParam		
Attribute Name	Type	Optional	Description
edit_cloud_system_team_action	EditCloudSystemTeamActionParamType	No	Select one of the Edit Options to modify the specified Cloud System Network Team

Object	EditCloudSystemTeamActionParamType		
Attribute Name	Type	Optional	Description

edit_team_settings	EditTeamSettingsParamType	YES	defines the new ip settings for the Cloud System Team.

Object	EditTeamSettingsParamType		
Attribute Name	Type	Optional	Description
ip_address	String	NO	The IP Address of the Team
prefix	Interger	NO	The Netmask Prefix of the Team
interface_id	Array[CloudSystemInterfaceIdType]	NO	The New List of Interface ids to be associated with the NIC Team

Response Parameters :

SYNCHRONOUS:

Object	CloudSystemTeam		
Attribute Name	Type	Optional	Description
<CloudSystemInterface> is detailed in GET (Object)			

ASYNCHRONOUS:

Object	job		
Attribute Name	Type	Optional	Description
<Job> is detailed in the Appendix.			

2.6.11.3.1 Sample JSON request

```
{
  " edit_cloud_system_team_action": {
```

```
    " edit_team_settings": {
        “ip_address”:”10.10.10.10”,
        “prefix”:23
        “interface_id: [
             “eth1”,
            “eth2”
        ]
    }
  }
}
```

2.6.11.4 DELETE

Resource	team
Request Type:	DELETE
New Endpoint Description:	Deletes a Cloud System Network Team
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL:	…/92/system/symmetrix/{symmetrixId}/cloud/networking/team/{team_id}
User Roles:	Administrator/Storage Administrator
Prerequisite:	Not applicable
Path Parameters:	

Attribute Name	Type	Optional	Description
symmetrixId	String	No	Unique identifier of the symmetrix
team_id	String	No	The Cloud System Network Team Id

Query Parameters:	Not applicable
Request Parameters:	Not applicable
Response Parameters:	No content

2.6.12 audit_log_record

2.6.12.1 GET (List)

<table>
<tr><td>Resource</td><td>audit_log_record</td></tr>
<tr><td>Request Type:</td><td>GET</td></tr>
<tr><td>New Endpoint Description:</td><td>List Audit log records with a subset of attributes for a symmetrix</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>… /92/system/symmetrix/{symmetrixId}/audit_log_record</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>Unique identifier of the symmetrix</td></tr>
</table>
</td></tr>
<tr><td>Query Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr>
<tr><td>application_id</td><td>String</td><td>Optional value that filters returned list to display Audit Log Entries that contain the specified application Id only, for example, equal to "applicationr_id=UNIVMAX"</td></tr>
<tr><td>record_id</td><td>long</td><td>Optional value that filters returned list to display Audit Log Entries that have a record_id equal to ("record_id=123456")</td></tr>
<tr><td>username</td><td>String</td><td>Optional value that filters returned list to display Audit Log Entries that contain the specified username only, for example, equal to "username=user1"</td></tr>
</table>
</td></tr>
</table>

	message	String	Optional value that filters returned list to display Audit Log Entries that contain the specified message only, for example, equal to or like" message =string"
	hostname	String	Optional value that filters returned list to display Audit Log Entries that contain the specified username only, for example, equal to " hostname =IRCO2SD30VM1"
	os_type	String	Optional value that filters returned list to display Audit Log Entries that contain the specified os_type only, for example, equal to " os_type =WinNT"
	vendor_id	String	Optional value that filters returned list to display Audit Log Entries that contain the specified vendor_id only, for example, equal to " vendor_id =EMC Corp"
	audit_class	String	Optional value that filters returned list to display Audit Log Entries that contain the specified audit_class only, for example, equal to " audit_class =Snap"
	action_code	String	Optional value that filters returned list to display Audit Log Entries that contain the specified action_code only, for example, equal to “action_code=SPPEstablish"
	activity_id	String	Optional value that filters returned list to display Audit Log Entries that contain the specified activity_id only, for example, equal to " activity_id =SEd813c3bbb5"
	entry_date	Long	Optional value that filters list between a range determined by values greater than milliseconds value entered and less than milliseconds value entered. Audit_log_record?entry_date>1572870639&entry_date<1572870650
Request Parameters:	Not applicable		

2.6.12.1.1 Sample JSON response


```
{
   "resultList": {
      "result": [
         {
            "record_id": 144071,
            "entry_date_string": "Wed Oct 2 2019 15:53:20",
            "entry_date_milliseconds": 1570029255317,
            "username": "H:irco2sd35vm11\\smc",
            "message": "Deleted job: 1569981151181 Raising Cloud Alert ID 313",
         }
 }
```

2.6.12.2 GET (Object)

<table>
<tr><td>Resource</td><td>audit_log_record</td></tr>
<tr><td>Request Type:</td><td>GET</td></tr>
<tr><td>New Endpoint Description:</td><td>Gets a single audit log record’s details</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>… /92/system/symmetrix/{symmetrixId}/audit_log_record/{record_id}</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>Unique identifier of the symmetrix</td></tr>
<tr><td>recordId</td><td>Long</td><td>Unique identifier for a record Id</td></tr>
</table>
</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
</table>

2.6.12.2.1 Sample JSON response


```
{
   "resultList": {
      "result": [
         {
            "record_id": 144071,
            "entry_date_string": "Wed Oct 2 2019 15:53:20",
            "entry_date_milliseconds": 1570029255317,
            "username": "H:irco2sd35vm11\\smc",
            "message": "Deleted job: 1569981151181 Raising Cloud Alert ID 313",
            "hostname": "IRCO2SD35VM1",
            "client_host": "Delete",
            "os_type": "WinNT",
            " os_revision ": "10.0.14393",
```

```
            "vendor_id": "EMC Corp",
            "application_id": "UNIVMAX",
            "application_version": "9.2.0.923",
            "api_library": "SDK",
            "api_version": "X9.2.0.621 (Edit Level: 2601)",
            "function_class": "N/A",
            "audit_class": "Job",
            "action_code": "Delete",
            "task_id": 7996,
            "activity_id": "SE6d33057dfd",
            "process_id": 5956,
            "offset_in_sequence": 1,
            "records_in_sequence": 1
        }
}
```


2.6.12.3 POST (EXPORT AUDIT LOG)

<table>
<tr><td>Resource</td><td>audit_log_record</td></tr>
<tr><td>Request Type:</td><td>POST</td></tr>
<tr><td>New Endpoint Description:</td><td>Exports Audit Log for a symmetrix</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>… /92/system/symmetrix/{symmetrixId}/audit_log_record/exportfile</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>Unique identifier of the symmetrix</td></tr>
</table>
</td></tr>
</table>

Query Parameters:	Not applicable
Request Parameters:	Not applicable
Response Parameters:	Object: Pdf file

2.6.12.3.1 Sample JSON Request

```
{
 "auditlogfilename": "auditlog_45"
}
```

2.6.13 job

2.6.13.1 GET (Object)

Resource	job
Request Type:	GET
New Endpoint Description:	Not applicable
Modify Endpoint Description:	**Added in this Release** Adding the following attribute and query param: • symmetrixId **Modified in this Release** In certain use cases this endpoint returned a 303 which re-directed the response to the resorce that was created during the job, this is no longer the expected behavior, instead the job will always be returned under non error conditions. **Removed in this Release** Not applicable

<table>
<tr><td></td><td></td></tr>
<tr><td>URL:</td><td>…/92/system/job/{jobId}</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>No change</td></tr>
<tr><td>Query Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>Optional value that filters returned list to display Jobs that have a specified symmetrixId example: equal to "symmetrixId=000197800111" or contains "symmetrixId=<like>111"</td></tr>
</table>
</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>
<table>
<tr><th>Object</th><th colspan="3">job</th></tr>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>Yes</td><td>The Jobs’ related symmetrix id</td></tr>
</table>
</td></tr>
</table>

2.6.13.2 GET (List)

<table>
<tr><td>Resource</td><td>symmetrix</td></tr>
<tr><td>Request Type:</td><td>GET</td></tr>
<tr><td>New Endpoint Description:</td><td>Not applicable</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Modified in this Release</th></tr>
</table>
</td></tr>
</table>

	In certain use cases this endpoint returned a 303 which re-directed the response to the resorce that was created during the job, this is no longer the expected behavior, instead the job will always be returned under non error conditions. Removed in this Release Not applicable
URL:	…/92/system/symmetrix/{symmetrixId}/job/{jobId}
User Roles:	Not applicable
Prerequisite:	Not applicable
Path Parameters:	Not applicable
Query Parameters:	Not applicable
Request Parameters:	Not applicable
Response Parameters:	Not applicable

2.6.14 Symmetrix

2.6.14.1 GET (Object)

Resource	symmetrix
Request Type:	GET
New Endpoint Description:	Not applicable
Modify Endpoint Description:	Added in this Release Adding the following attribute: • ucode_registered_build • service_level_rt_multiplier system_sized_property

	Modified in this Release
	Not applicable
	Removed in this Release
	Not applicable
URL:	…/92/system/symmetrix/{symmetrixId}
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable
Path Parameters:	No change
Query Parameters:	No change
Request Parameters:	Not applicable

Response Parameters:

Object	symmetrix		
Attribute Name	Type	Optional	Description
ucode_registered_build	int	Yes	Currently loaded Microcode Registered Build Number
service_level_rt_multiplier	string	Yes	The Service Level Response Time Multiplier that is currently set on the array
system_sized_property	SystemSizedPropertiesType Array	Yes	The Service Level Response Time Multiplier that is currently set on the array

Object	SystemSizedPropertiesType		
Attribute Name	Type	Optional	Description
srp_name	String	No	The SRP the system sized properties are being shown for
sized_fba_data_reduction_ratio	String	No	The FBA Data Reduction Ratio in the format “1.0:1”
sized_fba_capacity_tb	int	Yes	The sized FBA capacity. Shown only if SRP is FBA or Mixed
sized_ckd_capacity_tb	int	Yes	The sized CKD capacity. Shown only if SRP is CKD or Mixed

2.6.14.1.1 Sample JSON response

```
{
    …
    “service_level_rt_multiplier”: “Low”
    …
   "ucode_registered_build": 8
    …
"system_sized_property": [
    {
      "srp_name": "SRP_1",
      "sized_fba_data_reduction_ratio": "3.0:1",
      "sized_fba_capacity_tb": 102
    },
    {
      "sized_ckd_capacity_tb": 4,
      "srp_name": "SRP_0x102",
      "sized_fba_data_reduction_ratio": "3.0:1",
      "sized_fba_capacity_tb": 3
    },
    {
      "srp_name": "SRP_33",
      "sized_fba_data_reduction_ratio": "1.0:1"
    }
  ]
}
```

2.6.14.1 PUT

Resource	EditSymmetrixParam
Request Type	PUT

New Endpoint Description: Not applicable

Modify Endpoint Description:

Added in this Release
EditSymmetrixActionParam.Set_Service_Level_Rt_Multiplier_Param
Modified in this Release
Not applicable
Removed in this Release
Not applicable

URL …/92/system/symmetrix/{symmetrixId}

User Roles StorageAdmin

Prerequisite Not applicable

Path Parameters

Attribute Name	Type	Description
symmetrixId	String	Unique identifier of the symmetrix

Query Parameters Not applicable

Request Parameters

Object	editSymmetrixActionParam		
Name	**Type**	**Optional**	**Description**
set_service_level_rt_multiplier_param	SetServiceLevelRTMultiplierParamType	Yes	The action to be performed.

Object	set_service_level_rt_multiplier_param		
Name	**Type**	**Optional**	**Description**
service_level_rt_multiplier_param	ServiceLevelRTMultiplierParamType	No	Populated for action to add ports

Object	service_level_rt_multiplier_param		
Name	**Type**	**Optional**	**Description**
None	string	No	No RT Mulitplier Delay
Low	string	No	Low RT Mulitplier Delay
Default	string	No	Default RT Mulitplier Delay

Response Parameters: Symmetrix

2.6.14.1.1 Sample JSON Request

```
{
   "editSymmetrixActionParam": {
      "set_service_level_rt_multiplier_param": {
         "service_level_rt_multiplier_param": "Default"
      }
   }
}
```

2.7 Updates in 9.2.1

2.7.1 Common

- Starting with the 9.2.1 release of the public Unisphere for PowerMax REST API a 503 Error Respose Code will be returned in instances where the REST API Server is unable to fulfill the user request due to a temporary server overload, resource issue, or other related issue.

2.7.2 SLO Provisioning

2.7.2.1 symmetrix

2.7.2.1.1 GET (LIST)

Resource	symmetrix
Request Type:	GET
New Endpoint Description:	Not applicable
Modify Endpoint Description:	**Added in this Release** Added the following query params: • unreducible_data • reducible_data • deduplication_and_compression_savings • pattern_detection_savings • drr_on_reducible_only_to_one **Modified in this Release** Not applicable

	Removed in this Release Not applicable
URL:	…/92/sloprovisioning/symmetrix/
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable
Path Parameters:	Not applicable

Query Parameters:

Attribute Name	Type	Description
unreducible_data	String	Optional value that filters returned list to display symmetrix ids that contain greater than(">1"), Less than("<1") or equal to the specified unreducible_data
reducible_data	String	Optional value that filters returned list to display symmetrix ids that contain greater than(">1"), Less than("<1") or equal to the specified reducible_data
deduplication_and_compression_savings	String	Optional value that filters returned list to display symmetrix ids that contain greater than(">1"), Less than("<1") or equal to the specified deduplication_and_compression_savings
pattern_detection_savings	String	Optional value that filters returned list to display symmetrix ids that contain greater than(">1"), Less than("<1") or equal to the specified pattern_detection_savings
drr_on_reducible_only_to_one	String	Optional value that filters returned list to display symmetrix ids that contain greater than(">1"), Less than("<1") or equal to the specified drr_on_reducible_only_to_one

Request Parameters:	Not applicable
Response Parameters:	Not applicable

2.7.2.1.2 GET (Object)

<table>
<tr><td>Resource</td><td>symmetrix</td></tr>
<tr><td>Request Type:</td><td>GET</td></tr>
<tr><td>New Endpoint Description:</td><td>Not applicable</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Adding the following attribute:
unreducible_data_tb
reducible_data_tb
deduplication_and_compression_savings_tb
pattern_detection_savings_tb
drr_on_reducible_only_to_one</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>…/92/sloprovisioning/symmetrix/{symmetrixId}/</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>No change</td></tr>
<tr><td>Query Parameters:</td><td>No change</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>
<table>
<tr><th>Object</th><th colspan="3">System_efficiency</th></tr>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
</table>
</td></tr>
</table>

unreducible_data_tb	double	Y	Amount of unreducible data in TB
reducible_data_tb	double	Y	Amount of reducible data in TB
deduplication_and_compression_savings_tb	double	Y	Amount of savings in TB from Deduplication and Compression
pattern_detection_savings_tb	double	Y	Amount of savings in TB from Pattern detection
drr_on_reducible_only_to_one	double	Y	Data Reduction Ratio on Reducible data

2.7.2.1.2.1 ***Sample JSON response***

```
{
   …
"system_efficiency": {
    …
   "pattern_detection_savings_tb": 0.11,
   "drr_on_reducible_only_to_one": 4.1,
   "deduplication_and_compression_savings_tb": 0.95,
   "unreducible_data_tb": 0.02,
   "reducible_data_tb": 1.41
  },
   …
}
```

2.7.2.2 volume

2.7.2.2.1 GET (LIST)

Resource	volume
Request Type:	GET
New Endpoint Description:	Not applicable
Modify Endpoint Description:	**Added in this Release** Added the following query params: • unreducible_data_gb **Modified in this Release** Not applicable **Removed in this Release** Not applicable
URL:	…/92/sloprovisioning/symmetrix/{symmetrixId}/volume
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor

<table>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>Not applicable</td></tr>
<tr><td>Query Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr>
<tr><td>unreducible_data_gb</td><td>String</td><td>Optional value that filters returned list to display volume ids that contain greater than(">1"), Less than("<1") or equal to the specified unreducible_data_gb</td></tr>
</table>
</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>Not applicable</td></tr>
</table>

2.7.2.2.2 GET (Object)

<table>
<tr><td>Resource</td><td>volume</td></tr>
<tr><td>Request Type:</td><td>GET</td></tr>
<tr><td>New Endpoint Description:</td><td>Not applicable</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Adding the following attribute:
unreducible_data_gb</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>…/92/sloprovisioning/symmetrix/{symmetrixId}/volume/{volumeId}</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>No change</td></tr>
</table>

Query Parameters:	No change
Request Parameters:	Not applicable
Response Parameters:	

Object	srp		
Attribute Name	Type	Optional	Description
unreducible_data_gb	double	Y	Amount of unreducible data in GB

2.7.2.2.2.1 ***Sample JSON response***

```
{
   …
     “unreducible_data_gb”:100.0,
   …
}
```

2.7.2.3 storagegroup

2.7.2.3.1 GET (LIST)

Resource	storagegroup
Request Type:	GET
New Endpoint Description:	Not applicable
Modify Endpoint Description:	

Added in this Release
Added the following query params: • unreducible_data_gb
Modified in this Release
Not applicable

<table>
<tr><td></td><td>
<table>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>…/92/sloprovisioning/symmetrix/{symmetrixId}/storagegroup</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>Not applicable</td></tr>
<tr><td>Query Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr>
<tr><td>unreducible_data_gb</td><td>String</td><td>Optional value that filters returned list to display storage Group ids that contain greater than(">1"), Less than("<1") or equal to the specified unreducible_data_gb</td></tr>
</table>
</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>Not applicable</td></tr>
</table>

2.7.2.3.2 GET (Object)

<table>
<tr><td>Resource</td><td>storagegroup</td></tr>
<tr><td>Request Type:</td><td>GET</td></tr>
<tr><td>New Endpoint Description:</td><td>Not applicable</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Adding the following attribute:
unreducible_data_gb</td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>…/92/sloprovisioning/symmetrix/{symmetrixId}/storagegroup/{storageGroupId}</td></tr>
</table>

User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite :	Not applicable
Path Parameters:	No change
Query Parameters:	No change
Request Parameters:	Not applicable
Response Parameters:	

Object	storagegroup		
Attribute Name	Type	Optional	Description
unreducible_data_gb	double	Y	Amount of unreducible data in GB

2.7.2.3.2.1 ***Sample JSON response***

```
{
   …
     “unreducible_data_gb”:100.0,
   …
}
```

2.7.2.4 srp

2.7.2.4.1 GET (LIST)

Resource	srp
Request Type:	GET
New Endpoint Description :	Not applicable
Modify Endpoint Description :	Added in this Release

<table>
<tr><td></td><td>Added the following query params:
unreducible_data_tb
reducible_data_tb
deduplication_and_compression_savings_tb
pattern_detection_savings_tb
drr_on_reducible_only_to_one
<table><tr><th>Modified in this Release</th></tr><tr><td>Not applicable</td></tr><tr><th>Removed in this Release</th></tr><tr><td>Not applicable</td></tr></table></td></tr>
<tr><td>URL:</td><td>…/92/sloprovisioning/symmetrix/{symmetrixId}/srp</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>Not applicable</td></tr>
<tr><td>Query Parameters:</td><td><table><tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr><tr><td>unreducible_data_tb</td><td>String</td><td>Optional value that filters returned list to display symmetrix ids that contain greater than(">1"), Less than("<1") or equal to the specified unreducible_data_tb</td></tr><tr><td>reducible_data_tb</td><td>String</td><td>Optional value that filters returned list to display symmetrix ids that contain greater than(">1"), Less than("<1") or equal to the specified reducible_data_tb</td></tr><tr><td>deduplication_and_compression_savings_tb</td><td>String</td><td>Optional value that filters returned list to display symmetrix ids that contain greater than(">1"), Less than("<1") or equal to the specified deduplication_and_compression_savings_tb</td></tr><tr><td>pattern_detection_savings_tb</td><td>String</td><td>Optional value that filters returned list to display symmetrix ids that contain greater than(">1"), Less than("<1") or equal to the specified pattern_detection_savings_tb</td></tr></table></td></tr>
</table>

	drr_on_reducible_only_to_one	String	Optional value that filters returned list to display symmetrix ids that contain greater than(">1"), Less than("<1") or equal to the specified drr_on_reducible_only_to_one
Request Parameters:	Not applicable		
Response Parameters:	Not applicable		

2.7.2.4.2 GET (Object)

Resource	srp
Request Type:	GET
New Endpoint Description:	Not applicable
Modify Endpoint Description:	Added in this Release Adding the following attribute: • unreducible_data_tb • reducible_data_tb • deduplication_and_compression_savings_tb • pattern_detection_savings_tb • drr_on_reducible_only_to_one Modified in this Release Not applicable Removed in this Release Not applicable
URL:	…/92/sloprovisioning/symmetrix/{symmetrixId}/srp/{srpId}
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable
Path Parameters:	No change

Query Parameters:	No change
Request Parameters:	Not applicable

Response Parameters:

Object	srp		
Attribute Name	Type	Optional	Description
unreducible_data_tb	double	Y	Amount of unreducible data in TB
reducible_data_tb	double	Y	Amount of reducible data in TB
deduplication_and_compression_savings_tb	double	Y	Amount of savings in TB from Deduplication and Compression
pattern_detection_savings_tb	double	Y	Amount of savings in TB from Pattern detection

	drr_on_reducible_only_to_one	double	Y	Data Reduction Ratio on Reducible data

2.7.2.4.2.1 ***Sample JSON response***

```
{
   …
"system_efficiency": {
    …
   "pattern_detection_savings_tb": 0.11,
   "drr_on_reducible_only_to_one": 4.1,
   "deduplication_and_compression_savings_tb": 0.95,
   "unreducible_data_tb": 0.02,
   "reducible_data_tb": 1.41
  },
   …
}
```

2.7.2.5 storage_group_demand_report

2.7.2.5.1 GET (Object)

Resource	storage_group_demand_report
Request Type:	GET
New Endpoint Description:	Not applicable
Modify Endpoint Description:	Added in this Release Adding the following attribute: • unreducible_data_used_gb • unreducible_snapshot_used_gb Modified in this Release

	Not applicable Removed in this Release Not applicable
URL:	…/92/sloprovisioning/symmetrix/{symmetrixId}/srp/{srpId}/storage_group_demand_report
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable
Path Parameters:	No change
Query Parameters:	No change
Request Parameters:	Not applicable

Response Parameters:

Object	srp		
Attribute Name	Type	Optional	Description
unreducible_data_used_gb	double	Y	Amount of unreducible data in GB
unreducible_snapshot_used_gb	double	Y	Amount of reducible snapshot data in GB

2.7.2.5.1.1 ***Sample JSON response***

```
{
   …
     “unreducible_data_used_gb”:100.0,
     “unreducible_snapshot_used_gb”:100.0,
```

```
    …
}
```

2.7.3 System

2.7.3.1 director/port

2.7.3.1.1 GET (LIST)

Resource	port
Request Type:	GET
New Endpoint Description:	Not applicable
Modify Endpoint Description:	**Added in this Release** Added the following query params: • rx_power_level_mw • tx_power_level_mw • power_levels_last_sampled_date_milliseconds • mac_address **Modified in this Release** Not applicable **Removed in this Release** Not applicable
URL:	…/92/system/symmetrix/{symmetrixId}/director/{directorId}/port/
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable
Path Parameters:	Attribute Name / Type / Description symmetrixId / String / Unique identifier of the symmetrix DirectorId / String / Unique identifier of the director

<table>
<tr><td>Query Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr>
<tr><td>rx_power_level_mw</td><td>String</td><td>Optional value that filters returned list to display port ids that contain greater than(">1"), Less than("<1") or equal to the specified rx_power_level_mw</td></tr>
<tr><td>tx_power_level_mw</td><td>String</td><td>Optional value that filters returned list to display port ids that contain greater than(">1"), Less than("<1") or equal to the specified tx_power_level_mw</td></tr>
<tr><td>power_levels_last_sampled_date_milliseconds</td><td>String</td><td>Optional value that filters returned list to display Port names that have a specified registration date in milliseconds e.g. equal to "power_levels_last_sampled_date_milliseconds=123", greater than "power_levels_last_sampled_date_milliseconds=>123" or Less than "power_levels_last_sampled_date_milliseconds=<123"</td></tr>
<tr><td>mac_address</td><td>String</td><td>Optional value that filters returned list to display Port names that have a specified mac_address associated e.g. equal to "mac_address=00:60:48:23:fb:1b" or contains "mac_address=< like>23:fb:1b"</td></tr>
</table>
</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>Not applicable</td></tr>
</table>

2.7.3.1.2 GET (Object)

<table>
<tr><td>Resource</td><td>port</td></tr>
<tr><td>Request Type:</td><td>GET</td></tr>
<tr><td>New Endpoint Description:</td><td>Not applicable</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td>Adding the following attribute:</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><td></td><td>

rx_power_level_mw
tx_power_level_mw
power_levels_last_sampled_date_milliseconds
mac_address

<table>
<tr><th>Modified in this Release</th></tr>
<tr><td>Not applicable</td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>…/92/system/symmetrix/{symmetrixId}/director/{directorId}/port/{portId}</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>No change</td></tr>
<tr><td>Query Parameters:</td><td>No change</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>
<table>
<tr><th>Object</th><td colspan="3">port</td></tr>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>rx_power_level_mw</td><td>Double</td><td>Yes</td><td>The Receive Light Power level of the port in mW, (0 – 2)
Note: value maybe converted to E notation e.g. 0.0007 --> 7.0E-4</td></tr>
</table>
</td></tr>
</table>

tx_power_level_mw	Double	Yes	The Transmit Light Power Level of the Port in mW, (0 – 2) Note: value maybe converted to E notation e.g. 0.0007 --> 7.0E-4
power_levels_last_sampled_date_milliseconds	Long	Yes	The date in milliseconds since Epoch of the last time the power levels were recorded (sampled) , Power Levels are sampled ever 15 minutes.
mac_address	String	Yes	The port MAC Address

2.7.3.1.2.1 ***Sample JSON response***

```
{
   …
   “rx_power_level_mw”: 0.5269,
   “tx_power_level_mw”: 0.5269,
   “power_levels_last_sampled_date_milliseconds”: 1600764191639,
   “mac_address”: “00:5b:33:23:1a”
   …
}
```

2.7.3.2 cybersecIq

2.7.3.2.1 GET (Object)

<table>
<tr><td>Resource</td><td>cybersecIq</td></tr>
<tr><td>Request Type:</td><td>GET</td></tr>
<tr><td>New Endpoint Description:</td><td>New rest endpoint to get CyberSecIQ Data collection details</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><td>Added in this Release</td></tr>
<tr><td>Not applicable</td></tr>
<tr><td>Modified in this Release</td></tr>
<tr><td>Not applicable</td></tr>
<tr><td>Removed in this Release</td></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>… /92/system/cybersecIq</td></tr>
<tr><td>User Roles:</td><td>Administrator/ StorageAdministrator/ Security Administrator/ Monitor/ Auditor/ Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>Not applicable</td></tr>
<tr><td>Query Parameters:</td><td>Not applicable</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>
<table>
<tr><td>Object</td><td colspan="3">cybersecIq_details</td></tr>
<tr><td>Attribute Name</td><td>Type</td><td>Optional</td><td>Description</td></tr>
<tr><td>unisphere_registered</td><td>boolean</td><td>No</td><td>Details if Unisphere is registered or not</td></tr>
<tr><td>send_data_to_cloudiq</td><td>boolean</td><td>No</td><td>Details if Unisphere is sending data to CloudIQ</td></tr>
<tr><td>send_data_to_cybersecIq</td><td>boolean</td><td>No</td><td>Details if Unisphere is sending data to CyberSecIQ</td></tr>
</table>
</td></tr>
</table>

2.7.3.2.1.1 Sample JSON response

```
{
    "unisphere_registered": true,
```

```
    "send_data_to_cloudiq": true
    "send_data_to_cyberseciq": true
}
```

2.7.3.2.2 **PUT**

<table>
<tr><td>Resource</td><td>cyberseciq</td></tr>
<tr><td>Request Type:</td><td>PUT</td></tr>
<tr><td>New Endpoint Description:</td><td>New rest endpoint to set CyberSecIQ Data collection</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><td>Added in this Release</td></tr>
<tr><td>Not applicable</td></tr>
<tr><td>Modified in this Release</td></tr>
<tr><td>Not applicable</td></tr>
<tr><td>Removed in this Release</td></tr>
<tr><td>Not applicable</td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>… /92/system/cyberseciq</td></tr>
<tr><td>User Roles:</td><td>Administrator/ StorageAdministrator/ Security Administrator/ Monitor/ Auditor/ Performance Monitor</td></tr>
<tr><td>Prerequisite:</td><td>Direct Connection to ESRS Gateway
Data is being sent to CloudIQ</td></tr>
<tr><td>Path Parameters:</td><td>Not applicable</td></tr>
<tr><td>Query Parameters:</td><td>Not applicable</td></tr>
<tr><td>Request Parameters:</td><td>
<table>
<tr><td>Object</td><td colspan="3">SetCyberseciqDetailsParam</td></tr>
<tr><td>Attribute Name</td><td>Type</td><td>Optional</td><td>Description</td></tr>
<tr><td>send_data_to_cyberseciq</td><td>boolean</td><td>No</td><td>Details if Unisphere is sending data to CyberSecIQ</td></tr>
</table>
</td></tr>
<tr><td>Response Parameters:</td><td>
<table>
<tr><td>Object</td><td colspan="3">cyberseciq_details</td></tr>
<tr><td>Attribute Name</td><td>Type</td><td>Optional</td><td>Description</td></tr>
<tr><td>unisphere_registered</td><td>boolean</td><td>No</td><td>Details if Unisphere is registered or not</td></tr>
</table>
</td></tr>
</table>

	send_data_to_cloudiq	boolean	No	Details if Unisphere is sending data to CloudIQ
	send_data_to_cyberseciq	boolean	No	Details if Unisphere is sending data to CyberSecIQ

2.7.3.2.2.1 Sample JSON response

```
{
   "unisphere_registered": true,
   "send_data_to_cloudiq": true
   "send_data_to_cyberseciq": true
}
```

2.7.3.3 local_user

2.7.3.3.1 **PUT**

Resource	local_user
Request Type:	PUT
New Endpoint Description:	New rest endpoint to change a local user's password
Modify Endpoint Description:	Added in this Release Not applicable Modified in this Release Not applicable Removed in this Release Not applicable
URL:	… /92/system/local_user
User Roles:	Administrator/ StorageAdministrator/ Security Administrator/ Monitor/ Auditor/ Performance Monitor
Prerequisite:	Not applicable
Path Parameters:	Not applicable
Query Parameters:	Not applicable

Request Parameters:

Object	UpdateLocalUsers		
Attribute Name	Type	Optional	Description
username	string	No	Local user's username
action	LocalUsersAction	No	Action to be performed
set_password	LocalUsersSetPasswordParam	Yes	Param to set password

Object	LocalUsersAction
Attribute Name	Description
SetPassword	Set Password Action

Object	LocalUsersSetPasswordParam		
Attribute Name	Type	Optional	Description
current_password	string	No	Current password
new_password	string	No	New Password

Response Parameters:

*2.7.3.3.1.1 **Sample JSON request***

```
{
   "username":"user",
   "action" : "SetPassword",
   "set_password" : {
      "current_password" : "password",
      "new_password": "password1"
   }
}
```

2.7.3.3.1.2 Response Codes

Response status code	Description
204	OK. The password has been changed successfully.

400	Returned if the username, current password or new password are not provided, or the current password is the same text as the new password.
500	Returned if a problem occurred when changing the password for the provided user.

3 Appendix

3.1 Basic Authentication

Unisphere for VMAX REST API authorization is implemented using HTTP Basic Access Authentication. Basic Access Authentication allows the user to send an unencrypted Base64-encoded string that contains a username and password to the Unisphere for VMAX server.

Anonymous requests for a protected resource are rejected with a 401 (Access Denied) status code.

Known requests are validated against the existing Unisphere for VMAX users:

- Administrator
- Storage administrator
- Security administrator
- Monitor
- Auditor
- Performance monitor

3.2 Request Types

3.2.1 GET (Object)

3.2.1.1 Sample

Resource	storagegroup
Request Type:	GET
New Endpoint Description:	Gets a storage group Object
Modify Endpoint Description:	Added in this Release Modified in this Release Removed in this Release

URL:	… /84/sloprovisioning/symmetrix/{symmetrixId}/storagegroup/{storagegroupId}
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable
Path Parameters:	

Attribute Name	Type	Optional	Description
symmetrixId	String	No	Unique identifier of the symmetrix
storageGroupId	String	No	The Storage Group Name

Query Parameters: Not applicable

Request Parameters: Not applicable

Response Parameters:

Object	storageGroup		
Attribute Name	**Type**	**Optional**	**Description**
storageGroupId	String	No	Unique identifier of the storage group instance
slo	String	Yes	SLO that is associated with the storage group
srp	String	Yes	SRP that is associated with the storage group
hostIOLimit	HostIOLimitType	Yes	SRP that is associated with the storage group
And so on			

Object	HostIOLimitType		
Attribute Name	**Type**	**Optional**	**Description**
host_io_limit_mb_sec	String	No	The Host MB/s limit for the storage group
host_io_limit_io_sec	String	Yes	The Host IO/s limit for the storage group
dynamicDistribution	String	Yes	The dynamic distribution type, which can have "never", "always", or "on_failure" values

3.2.1.2 Response Codes

Response status code	Description
200	OK. The request is successful
401	Returned if an incorrect username, password, or both are specified
403	Returned if the specified username and password are correct, but the user is not authorized to make this request
404	Returned if the storage group [storageGroupId] on the Symmetrix [SymmetrixId] system cannot be found.
500	Returned if a problem occurred when the system was retrieving the <STORAGE_GROUP> resource: <STORAGE_GROUP_Name>
503	Returned if the server was unable to fulfill the request

3.2.1.3 Error codes:

Future work

3.2.2 GET (List)

3.2.2.1 Sample

Resource	storagegroup
Request Type:	GET
New Endpoint Description:	Get a List of storage group names
Modify Endpoint Description:	Added in this Release Modified in this Release Removed in this Release
URL:	… /84/sloprovisioning/symmetrix/{symmetrixId}/storagegroup
User Roles:	Administrator/Storage Administrator/Security Administrator/Monitor/Auditor/Performance Monitor
Prerequisite:	Not applicable

Path Parameters:

Attribute Name	Type	Description
symmetrixId	String	Unique identifier of the symmetrix

Query Parameters:

Attribute Name	Type	Description

	storageGroupId	String	An optional value that filters the returned list so it displays storage group names that contain the specified volume names only, for example, equal to "storageGroupId=gold_SG" or contain "storageGroupId=< like>gold"
	sym_vol_name	String	An optional value that filters the returned list so it displays storage group names that contain the specified volumes names only, for example, equal to "sym_vol_name=00123" or contain "sym_vol_name=< like>12"
	And so on		

Request Parameters:	Not applicable

Response Parameters:

Object	ListStorageGroupResult		
Attribute Name	Type	Optional	Description
storageGroupId	String array	No	A list of storage group names

3.2.2.2 Response Codes

Response status code	Description
200	OK. The request is successful
401	Returned if an incorrect username, password, or both are specified
403	Returned if the specified username and password are correct, but the user is not authorized to make this request
500	Returned if a problem occurred when the system was retrieving the <STORAGE_GROUP> resource: <STORAGE_GROUP_Name>
503	Returned if the server was unable to fulfill the request

Exception:

If the list of objects does not include a symmetrixId in the URL, a 200 response code is returned with an array of authorized objects, for example, …84//sloprovisioning/symmetrix returns only the VMAXs that the user has permissions on.

3.2.2.3 Error codes:

Future work

3.2.3 POST

3.2.3.1 Sample

Resource	storagegroup
Request Type:	POST

<table>
<tr><td>New Endpoint Description:</td><td>Creates a storage group</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td></td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td></td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td></td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>… /84/sloprovisioning/symmetrix/{symmetrixId}/storagegroup</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>Unique identifier of the symmetrix</td></tr>
</table>
</td></tr>
<tr><td>Query Parameters:</td><td>Not applicable</td></tr>
<tr><td>Request Parameters:</td><td>
<table>
<tr><th>Object</th><td colspan="3">createStorageGroupParam
extends:
ConfigurationManagementParamType (See the Appendix for details.)</td></tr>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>storageGroupId</td><td>String</td><td>No</td><td>Unique identifier of the new storage group instance</td></tr>
<tr><td>create_empty_storage_group</td><td>boolean</td><td>Yes</td><td>An empty storage group is created if you set this</td></tr>
</table>
</td></tr>
</table>

			parameter to true. This parameter overrides any other create setting.
sloBasedStorageGroupParam	SloBasedStorageGroupParamType	Yes	This parameter is required to create a SLO-based storage group. Multiple occurrences create multiple child storage groups.
And so on			

Object	SloBasedStorageGroupParamType		
Attribute Name	Type	Optional	Description
sloId	String	No	The SLO that is to be associated with the storage group
volumeAttribute	VolumeAttributeType	No	Size of each volume that is to be added to the storage group
And so on			

Object	VolumeAttributeType		
Attribute Name	Type	Optional	Description
volume_size	String	No	The size of the volume
capacityUnit	String enum: CYL MB	No	Unit of capacity

<table>
<tr><td></td><td><table>
<tr><td></td><td>GB
TB</td><td></td><td></td></tr>
</table></td></tr>
<tr><td>Response Parameters:</td><td>SYNCHRONOUS:<table>
<tr><td>Object</td><td colspan="3">storageGroup</td></tr>
<tr><td>Attribute Name</td><td>Type</td><td>Optional</td><td>Description</td></tr>
<tr><td colspan="4"><Insert the noun or noun phrase that describes what is detailed> is detailed in the GET method.</td></tr>
</table>
ASYNCHRONOUS:<table>
<tr><td>Object</td><td colspan="3">job</td></tr>
<tr><td>Attribute Name</td><td>Type</td><td>Optional</td><td>Description</td></tr>
<tr><td colspan="4"><Insert the noun or noun phrase that describes what is detailed> is detailed in the Appendix.</td></tr>
</table></td></tr>
</table>

3.2.3.2 Response Codes

Response status code	Description
200	OK, returned when endpoint is used for operations that do NOT create a Resource using **SYNCHRONOUS** execution option.
201	Created, returned when Resource is created using **SYNCHRONOUS** execution option.
202	Accepted, returned when the request to create the resource is accepted using the **ASYNCHRONOUS** execution option.
400	Returned when an error occurs with the validation of the Request payload , for example:<STORAGE_GROUP> must be 1-64 alphanumeric, '_', or '-' characters
401	Returned if an incorrect username, password, or both are specified
403	Returned if the specified username and password are correct, but the user is not authorized to make this request
409	Returned if the requested <STORAGE_GROUP> resource already exists.
500	Returned if a problem occurred when the system was creating the <STORAGE_GROUP> resource: <STORAGE_GROUP_Name>
503	Returned if the server was unable to fulfill the request

3.2.3.3 Error codes:

Future work

3.2.3.4 Response Header

Asynchronous

The URL of the job

- Example: restapi/84/system/job/{jobId}

Synchronous:

The URL of the resource that is being created:

- Example:restapi/84/sloprovisioning/symmetrix/{symmetrixId}/storagegroup/{storageGroupId}

3.2.4 PUT

3.2.4.1 Sample

<table>
<tr><td>Resource</td><td>storagegroup</td></tr>
<tr><td>Request Type:</td><td>PUT</td></tr>
<tr><td>New Endpoint Description:</td><td>Modifies a storage group</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td></td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td></td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td></td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>… /84/sloprovisioning/symmetrix/{symmetrixId}/storagegroup/{storageGroupID}</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>Unique identifier of the symmetrix</td></tr>
<tr><td>storageGroupId</td><td>String</td><td>The Storage Group Name</td></tr>
</table>
</td></tr>
<tr><td>Query Parameters:</td><td>Not applicable</td></tr>
<tr><td>Request Parameters:</td><td>
<table>
<tr><th>Object</th><td>editStorageGroupParam
extends:
ConfigurationManagementParamType (See the Appendix for details.)</td></tr>
</table>
</td></tr>
</table>

Attribute Name	Type	Optional	Description
editStorageGroupActionParam	EditStorageGroupParamType	No	Select one of the Edit Options to modify the specified storage group

Object	EditStorageGroupParamType		
Attribute Name	Type	Optional	Description
renameStorageGroupParam	RenameStorageGroupParamType	Yes	Renames the specified storage group
editStorageGroupSLOParam	EditStorageGroupSLOParamType	Yes	Edit the SG SLO Applicable to stand-alone and child storage groups only
and so on.			

Object	RenameStorageGroupParamType		
Attribute Name	Type	Optional	Description
new_storage_Group_name	String	No	The New Storage Group Name.

Object	EditStorageGroupSLOParamType		
Attribute Name	Type	Optional	Description
sloId	String	No	The New SLO Selection

Response Parameters:	SYNCHRONOUS:

Object	storageGroup		
Attribute Name	Type	Optional	Description
<Insert the noun or noun phrase that describes what is detailed> is detailed in the GET method.			

ASYNCHRONOUS:

Object	job		
Attribute Name	Type	Optional	Description
<Insert the noun or noun phrase that describes what is detailed> is detailed in the Common Objects section.			

3.2.4.2 Response Codes

Response status code	Description
200	Ok, returned when Resource is edited using **SYNCHRONOUS** execution option.
202	Accepted, returned when the request to edit the resource is accepted using the **ASYNCHRONOUS** execution option. Note: In some rare circumstances, such as unmapping all volumes from a cu image, the cu image no longer exists; hence, there is no object or content to be returned. In these circumstances, a 204: No content will be returned after the job completes
400	Returned when an error occurs with the validation of the Request payload , for example:<STORAGE_GROUP> must be 1-64 alphanumeric, '_', or '-' characters
401	Returned if an incorrect username, password, or both are specified
403	Returned if the specified username and password are correct, but the user is not authorized to make this request
409	Returned if the requested <STORAGE_GROUP> resource already exists.
500	Returned if a problem occurred when the system was editing the <STORAGE_GROUP> resource: <STORAGE_GROUP_Name>
503	Returned if the server was unable to fulfill the request

3.2.4.3 Error codes:

Future work

3.2.4.4 Response Header:

Asynchronous:

The URL of the job

- For example, restapi/84/system/job/{jobId}

Synchronous:

A URI Header cannot be passed back in a 200(ok) Response

3.2.5 DELETE

3.2.5.1 Sample

<table>
<tr><td>Resource</td><td>storagegroup</td></tr>
<tr><td>Request Type:</td><td>DELETE</td></tr>
<tr><td>New Endpoint Description:</td><td>Deletes a storage group</td></tr>
<tr><td>Modify Endpoint Description:</td><td>
<table>
<tr><th>Added in this Release</th></tr>
<tr><td></td></tr>
<tr><th>Modified in this Release</th></tr>
<tr><td></td></tr>
<tr><th>Removed in this Release</th></tr>
<tr><td></td></tr>
</table>
</td></tr>
<tr><td>URL:</td><td>… /84/sloprovisioning/symmetrix/{symmetrixId}/storagegroup/{storagegroupId}</td></tr>
<tr><td>User Roles:</td><td>Administrator/Storage Administrator</td></tr>
<tr><td>Prerequisite:</td><td>Not applicable</td></tr>
<tr><td>Path Parameters:</td><td>
<table>
<tr><th>Attribute Name</th><th>Type</th><th>Optional</th><th>Description</th></tr>
<tr><td>symmetrixId</td><td>String</td><td>No</td><td>Unique identifier of the symmetrix</td></tr>
<tr><td>storageGroupId</td><td>String</td><td>No</td><td>The Storage Group Name</td></tr>
</table>
</td></tr>
<tr><td>Query Parameters:</td><td>Not applicable</td></tr>
<tr><td>Request Parameters:</td><td>Not applicable</td></tr>
<tr><td>Response Parameters:</td><td>No content</td></tr>
</table>

3.2.5.2 Response Codes:

Response status code	Description
204	No Content, returned when Resource is deleted successfully
401	Returned if an incorrect username, password, or both are specified
403	Returned if the specified username and password are correct, but the user is not authorized to make this request

404	Returned if the requested <STORAGE_GROUP> resource cannot be found
500	Returned if a problem occurred when the system was deleting the <STORAGE_GROUP> resource: <STORAGE_GROUP_Name>
503	Returned if the server was unable to fulfill the request

3.2.5.3 Error codes:

Future work

3.3 Common Objects

3.3.1 Iterator

An Iterator is a paged list of objects that is cached for a set period of time. Until the expiry time has been reached, the list of objects can be queried by page from the following locations:

…/restapi/common/Iterator/{iteratorId}

../restapi/common/Iterator/{iteratorId}/page?from=1050&to=1100

Object	iterator		
Attribute Name	Type	Optional	Description
id	String	No	Unique iterator identifier that is used to look up the result until the expiry time is reached
count	int	No	Total number of result objects
ExpirationTime	String	No	Unix timestamp that shows when the result will be removed from cache and no longer queryable
maxPageSize	int	No	Maximum number of results that is allowed per result list page
result	Object Array	No	Array of objects that is returned

3.3.1.1 Sample Object Array:

Object	Iterator that contains list of: daysToFullResultType		
Attribute Name	Type	Optional	Description
projectionDaysToFull	float	Yes	Number of days until the specified category will be full, - to a maximum of 180 days
ProjectionMaxProjectedData	float	Yes	Maximum number of days that will be projected
ProjectionCapacityTrend	float	Yes	Projected capacity

PercentUsedCapacity	double	Yes	Current percentage of used capacity
TotalPoolCapacityGB	double	Yes	Current total capacity
instanceId	String	No	The unique identifier of the returned category object

3.3.1.1.1 Sample JSON Response:


```
{
  "resultList": {
    "result": [
      {
        "ProjectionDaysToFull": 181,
        "ProjectionMaxProjectedData": 180,
        "ProjectionCapacityTrend": 0,
        "PercentUsedCapacity": 35.10973774908589,
        "TotalPoolCapacityGB": 45727.2,
        "instanceId": "000000000001"
      }
    ],
    "from": 1,
    "to": 1
  },
  "id": "31d90275-c8d3-42d2-ab98-0343a3b3b16e_0",
  "count": 1,
  "expirationTime": 1474474773510,
  "maxPageSize": 1000
}
```


3.3.2 Job

Object	job		
Attribute Name	Type	Optional	Description
jobId	String	No	Unique identifier of the job instance
name	String	Yes	Name of the job
status	JobStatus enum: CREATED SCHEDULED	No	Current status of the job

	RUNNING SUCCEEDED FAILED ABORTED UNKNOWN VALIDATING VALIDATED VALIDATE_FAILED INVALID RETRIEVING_PICTURE		
username	string	No	User who initiated the job
last_modified_date	String	Yes	Date the job was last modified
last_modified_date_milliseconds	String	Yes	Date the job was last modified in milliseconds
scheduled_date	String	No	Date the job was last scheduled
scheduled_date_milliseconds	String	Yes	Date the job was last scheduled in milliseconds
completed_date	String	Yes	Date the job was completed
completed_date_milliseconds	String	Yes	Date the job was last completed in milliseconds
task	TaskType	Yes	Current status of the job
resourceLink	String	Yes	URL of the object that is created, when applicable
result	String	Yes	Description of the job outcome, its failure or success

Object	TaskType		
Attribute Name	Type	Optional	Description
execution_order	int	Yes	Order of the task
description	String	Yes	Description of the task

3.3.2.1 Sample JSON Response:


```
{
  "jobId": "1476181321815",
  "name": "Create SAMPLE_REST_PG ",
  "status": " SUCCEEDED ",
  "username": "smc",
```

```
  "last_modified_date": "Oct-11-2016 11:22:01.817",
  "last_modified_date_milliseconds": 1476181321817,
  "scheduled_date": "",
  "scheduled_date_milliseconds": 0,
  "completed_date": " Oct-11-2016 11:22:01.817",
  "completed_date_milliseconds": 1476181321817,
  "task": [
   {
    "execution_order": 1,
    "description": "Create Port Group : SAMPLE_REST_PG"
   }
  ],
 "resourceLink":
https://IP_ADDRESS:8443/univmax/restapi/84/sloprovisioning/symmetrix/{symmetrixId}/portgroup/SAMPLE_REST_PG,
   "result": "SUCCEEDED - Started job execution on Fri 27 Jan 2017 07:57:59 AM GMT
Finished job execution on Fri 27 Jan 2017 07:58:49 AM GMT
Successfully created volume(s):  01076",

}
```


3.3.2.2 Response Codes:

Response status code	Description
200	Ok, returned when Resource is retrieved successfully
401	Returned if an incorrect username, password, or both are specified
403	Returned if the specified username and password are correct, but the user is not authorized to make this request
404	Returned if the requested resource cannot be found
500	Returned if a problem occurred when the system was retrieving the resource: <RESOURCE_NAME>
503	Returned if the server was unable to fulfill the request

3.3.2.3 Error codes:

Future work

3.3.3 Configuration Management Type

Object	ConfigurationManagementParamType		
Attribute Name	Type	Optional	Description
executionOption	String enum: SYNCHRONOUS	Yes	Indicates whether the requested

	ASYNCHRONOUS		configuration management request should run synchronously or asynchronously

3.4 Automatic Cleanup of REST API jobs

All asynchronous and synchronous configuration change requests that are made through the REST API create a Job resource. Over time, the number of completed jobs accumulates and, without a mechanism to delete those jobs via the REST API, system performance might be impacted.

Because REST API jobs can be considered temporary resources that are relevant only while jobs run and to preempt any performance degradation, an automatic REST API cleanup task could perform a valuable function by deleting completed jobs.

The following list provides the functional requirements for implementing the automatic cleanup of REST API jobs:

- Only jobs that are created for REST API requests are eligible for automatic deletion.
- A cleanup task runs at a default of every 24 hours and deletes any eligible REST API jobs.
- A REST API job is eligible for deletion if it meets the following requirements:
  - Its status is one of SUCCEEDED, FAILED, or ABORTED.
  - At least the 24-hour default has passed since the job completed.
- The minimum time that must pass before a job is eligible for deletion is specified in hours, with a default value of 24. The valid range of values is from 1 to 32,767. The interval is configurable, and a new configuration requires an offline change to become active.
- Automatic cleanup of REST API jobs is enabled by default, but it is configurable. Cleanup can be disabled, and any new configuration requires an offline change to become active.

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